Holographic RIS-Aided Reference Energy Detection for Low-Complexity Noncoherent Communications

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***Abstract*—This paper presents a comprehensive framework for signal recovery in holographic reconfigurable intelligent surface (RIS) receivers, where only amplitude information is available at the receiver. The proposed system employs a known reference wave to enable phase retrieval from intensity-only measurements. We formulate the complete signal model from transmission to reception, and propose a Gerchberg-Saxton algorithm to estimate the wave vector and complex channel gain. Building upon this, we develop a Gauss-Newton algorithm for efficient signal recovery to conduct real-time estimation of the symbols utilizing the property of rapid convergence. Simulation results validate the effectiveness of the proposed approach, showing accurate signal recovery under various signal-to-noise ratio conditions. The complete modeling framework and algorithm implementation details are provided to facilitate reproducibility and further research in RIS-based communication systems.**

***Index Terms*—Reconfigurable Intelligent Surface, holographic receiver, phase retrieval, signal recovery, amplitude-only measurements.**

I. Introduction

Reconfigurable intelligent surfaces (RISs) have emerged as a promising technology for next-generation wireless communication systems, owing to their ability to reshape the wireless propagation environment in a programmable and energy-efficient manner [1], [2]. By adaptively controlling the electromagnetic response of a large number of low-cost reflecting elements, RISs can provide additional degrees of freedom for coverage enhancement, interference management, and spatial signal manipulation. More recently, the functionality of RISs has been extended beyond passive beamforming toward active reception and signal processing, giving rise to the concept of holographic RIS receivers [3]. By exploiting densely deployed metasurface elements, holographic RIS receivers are expected to capture rich spatial information of impinging electromagnetic waves while maintaining much lower hardware complexity than conventional antenna arrays equipped with fully digital radio-frequency chains.

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Despite these advantages, a fundamental bottleneck of holographic RIS receivers lies in the acquisition of complex-valued signal information. Conventional coherent receivers rely on down-conversion, local oscillator synchronization, and in-phase/quadrature sampling to obtain both amplitude and phase information. In contrast, practical RIS-based or metasurface-based receivers may only have access to intensity or power measurements due to hardware, power, and integration constraints [4]. As a result, the received signal phase is not directly observable, and the recovery of complex-valued symbols from intensity-only measurements naturally leads to a phase retrieval problem, which has been extensively studied in optics and signal processing [5], [6].

A promising way to alleviate this ambiguity is to introduce a known reference wave before intensity detection. Inspired by optical holography, where a reference beam interferes with an object wave to encode both amplitude and phase information into an intensity pattern [7], a reference-assisted RIS receiver superimposes a known radio-frequency reference signal with the unknown received signal at the metasurface or receiver front end. The resulting square-law measurement contains not only the power of the unknown signal but also a reference-induced interference term. This interference term serves as a phase anchor and makes it possible to infer the hidden complex-valued signal from real-valued intensity observations.

[8] investigated the application of phase retrieval techniques for channel estimation in orthogonal frequency-division multiplexing (OFDM) systems, demonstrating that accurate channel estimation is possible even when only magnitude information is available, which shows the phase retrieval problems often appear in the communication area. [9] proposed an affine phase retrieval framework for wireless communication systems, where the received signal is modeled as an affine transformation of the transmitted signal, and developed algorithms for signal recovery under this model, where the problems formulation is very similar to the reference-assisted phase retrieval model in this paper.

However, existing studies on RIS-assisted reception and phase retrieval still leave several key questions insufficiently addressed. Most RIS receiver designs focus on enhancing the received signal power, improving beamforming gain, or reducing hardware complexity, while the intrinsic identifiability of complex symbols under intensity-only measurements has not been fully characterized. In particular, when the receiver observes only the squared magnitude of the superimposed field, different transmitted symbols may produce identical or nearly identical intensity patterns. In such cases, no detector, including maximum-likelihood detection or iterative phase retrieval algorithms, can reliably distinguish the transmitted symbols. Therefore, the central issue is not only how to design an efficient recovery algorithm, but also whether the transmitted symbols are uniquely encoded in the intensity domain in the first place.

This issue becomes more critical in holographic RIS receivers because the measured intensity pattern is jointly determined by multiple coupled components, including the reference wave, the pulse-shaping or sampling operation, and the RIS array manifold. If the effective matrix maps two different symbol vectors to the same field response, the corresponding intensity measurements remain indistinguishable regardless of how the detector is implemented. Likewise, even when the effective field responses are different, an improperly designed reference wave may fail to project their phase difference into the observable intensity domain.

Consequently, the lack of a systematic identifiability analysis limits the theoretical understanding and practical design of reference-assisted holographic RIS receivers. Existing phase retrieval methods often assume generic random measurements or focus on continuous signal recovery, whereas communication symbols belong to finite constellations with specific algebraic structures. This discrete nature creates both opportunities and challenges: on one hand, exact recovery may be possible under weaker conditions than those required for general continuous phase retrieval; on the other hand, constellation-dependent ambiguities may still arise if the reference wave and the RIS measurement structure are not properly designed. Therefore, it is essential to investigate how the reference wave, the shaping matrix, and the RIS array manifold jointly determine the uniqueness and stability of symbol recovery.

In this work, we address this problem by developing a reference-assisted phase retrieval framework for holographic RIS receivers. Instead of treating the reference signal merely as an auxiliary hardware component, we characterize its role as a phase-anchoring mechanism that converts hidden phase information into measurable intensity variations. We analyze the discrete identifiability condition of transmitted symbols under the reference-assisted intensity observation model and show that successful recovery requires two complementary properties: the effective RIS measurement matrix must preserve the differences between constellation points, and the reference wave must separate these differences in the intensity domain. Based on this insight, we further introduce an energy-domain minimum distance metric to quantify the distinguishability of different transmitted symbols after square-law detection. This metric provides a direct link between reference-wave design, RIS measurement diversity, and symbol detection performance.

The main contributions of this paper are summarized as follows:

- We establish a reference-assisted holographic RIS receiver model, in which complex-valued communication symbols are recovered from intensity-only measurements through the interference between the unknown signal and a known reference wave.
- We derive discrete identifiability conditions for reference-assisted phase retrieval over finite constellations and reveal the joint impact of the reference wave, the pulse-shaping matrix, and the RIS array manifold on symbol uniqueness.
- We introduce an energy-domain minimum distance metric to characterize the stability of symbol recovery and to guide reference-wave and RIS measurement design.
- We develop and compare maximum-likelihood and low-complexity phase retrieval based detectors, demonstrating that proper reference-wave diversity significantly improves symbol distinguishability and detection performance.

II. System Model

We consider a holographic RIS receiver system in which L single-antenna users transmit complex baseband symbols to an RIS composed of K reflecting elements. The RIS senses only the intensity of the interference pattern formed by the received signal and a known reference wave at each of its elements. The known reference signal is generated by a local oscillator [10]. The goal is to estimate the complex channel parameters and degree of arrival (DOA) angles, and subsequently recover the transmitted symbols utilizing these channel information.

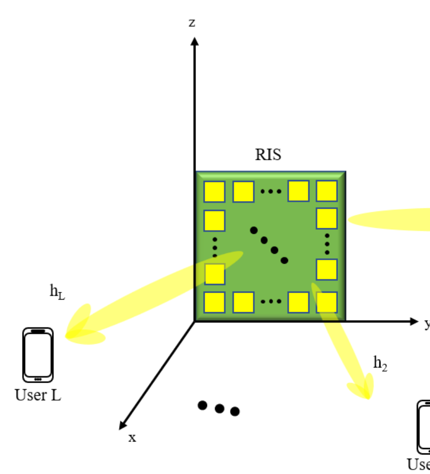

Fig. 1: System model of holographic RIS receiver with amplitude-only measurements.

A. Transmitter Model

Consider an uplink transmission system with L single-antenna users. The information-bearing symbols transmitted by the l-th user within one signaling block are collected as

$\mathbf{s}_l = [s_{l,0}, s_{l,1}, \ldots, s_{l,K-1}]^T \in \mathcal{S}^K$, where K denotes the number of symbols in the considered block, $s_{l,k}$ represents the k-th transmitted symbol of user l, and $\mathcal{S}$ denotes the modulation constellation. And the constellation symbols are assumed to be normalized as $\mathbb{E}\{|s_{l,k}|^2\} = 1$.

After pulse shaping, the complex baseband transmit waveform of the l-th user is given by

$$x_l(t) = \sqrt{P_l}\left(\sum_{k=0}^{K-1} s_{l,k} p(t - kT_s) + \xi_l q(t)\right), \tag{1}$$

where P_l is the average transmit power of user l, $p(t)$ is the pulse-shaping filter, T_s is the symbol duration, $\xi_l \in \mathbb{C}$ is a deterministic pilot or bias coefficient, and $q(t)$ is a known pilot waveform. The deterministic term $\xi_l q(t)$ is optional and is introduced to allow a nonzero-mean transmitted waveform when required by the subsequent reference-assisted recovery process. If no transmitted pilot or bias is used, we set $\xi_l = 0$.

The corresponding real-valued passband signal is expressed as

$$x_l^{\mathrm{RF}}(t) = \sqrt{2}\Re\left\{x_l(t)e^{j(2\pi f_c t + \varphi_l)}\right\}, \quad l = 1, 2, \ldots, L, \tag{2}$$

where f_c is the carrier frequency and φ_l denotes the initial carrier phase offset of the l-th user. Let

$$x_l(t) = x_{l,I}(t) + jx_{l,Q}(t), \tag{3}$$

where $x_{l,I}(t)$ and $x_{l,Q}(t)$ are the in-phase and quadrature components of the complex baseband waveform, respectively. Then (2) can be equivalently written as

$$\begin{aligned} x_l^{\mathrm{RF}}(t) = \sqrt{2}\big[&x_{l,I}(t)\cos(2\pi f_c t + \varphi_l) \\ &- x_{l,Q}(t)\sin(2\pi f_c t + \varphi_l)\big]. \end{aligned} \tag{4}$$

For compact notation, the multi-user symbol vector at symbol time k is defined as $\mathbf{s}_k = [s_{1,k}, s_{2,k}, \ldots, s_{L,k}]^T \in \mathcal{S}^L$, and the multi-user complex baseband waveform vector as

$$\mathbf{x}(t) = [x_1(t), x_2(t), \ldots, x_L(t)]^T \in \mathbb{C}^L. \tag{5}$$

Using (1), the multi-user baseband waveform can be written as

$$\mathbf{x}(t) = \mathbf{D}_P\left(\sum_{k=0}^{K-1} \mathbf{s}_k p(t - kT_s) + \boldsymbol{\xi} q(t)\right), \tag{6}$$

where $\mathbf{D}_P = \mathrm{diag}(\sqrt{P_1}, \sqrt{P_2}, \ldots, \sqrt{P_L})$ is the transmit-power scaling matrix, and

$$\boldsymbol{\xi} = [\xi_1, \xi_2, \ldots, \xi_L]^T \in \mathbb{C}^L \tag{7}$$

collects the deterministic pilot or bias coefficients of all users.

To facilitate the subsequent symbol-level analysis, we further introduce an equivalent matrix representation for the pulse-shaping operation. Consider N_t baseband observation instants $\{t_n\}_{n=0}^{N_t-1}$ within the signaling block. For the l-th user, define the sampled baseband transmit vector as

$$\tilde{\mathbf{x}}_l = [x_l(t_0), x_l(t_1), \ldots, x_l(t_{N_t-1})]^T \in \mathbb{C}^{N_t}. \tag{8}$$

From (1), we obtain

$$\tilde{\mathbf{x}}_l = \sqrt{P_l}\left(\mathbf{P}\mathbf{s}_l + \xi_l \mathbf{q}\right), \tag{9}$$

where $\mathbf{P} \in \mathbb{C}^{N_t \times K}$ is the pulse-shaping matrix with entries

$$[\mathbf{P}]_{n,k} = p(t_n - kT_s), \quad n = 0, 1, \ldots, N_t - 1, \quad k = 0, 1, \ldots, K-1, \tag{10}$$

and

$$\mathbf{q} = [q(t_0), q(t_1), \ldots, q(t_{N_t-1})]^T \in \mathbb{C}^{N_t} \tag{11}$$

is the sampled pilot waveform.

Stacking the sampled baseband transmit vectors of all users gives

$$\tilde{\mathbf{x}} = [\tilde{\mathbf{x}}_1^T, \tilde{\mathbf{x}}_2^T, \ldots, \tilde{\mathbf{x}}_L^T]^T \in \mathbb{C}^{LN_t}. \tag{12}$$

Similarly, define the stacked transmitted symbol vector as

$$\mathbf{s} = [\mathbf{s}_1^T, \mathbf{s}_2^T, \ldots, \mathbf{s}_L^T]^T \in \mathcal{S}^{LK}. \tag{13}$$

Then the sampled multi-user transmit waveform can be represented as

$$\tilde{\mathbf{x}} = (\mathbf{D}_P \otimes \mathbf{P})\mathbf{s} + (\mathbf{D}_P\boldsymbol{\xi}) \otimes \mathbf{q}, \tag{14}$$

where $\otimes$ denotes the Kronecker product. For notational simplicity, define

$$\mathbf{P}_{\mathrm{tx}} = \mathbf{D}_P \otimes \mathbf{P} \in \mathbb{C}^{LN_t \times LK} \tag{15}$$

and

$$\mathbf{x}_0 = (\mathbf{D}_P\boldsymbol{\xi}) \otimes \mathbf{q} \in \mathbb{C}^{LN_t}. \tag{16}$$

Thus, the transmitter-side equivalent discrete model is compactly written as

$$\tilde{\mathbf{x}} = \mathbf{P}_{\mathrm{tx}}\mathbf{s} + \mathbf{x}_0. \tag{17}$$

The matrix $\mathbf{P}_{\mathrm{tx}}$ characterizes the linear mapping from the transmitted constellation symbols to the sampled pulse-shaped waveforms. This representation allows the subsequent receiver model to incorporate pulse shaping, spatial propagation, and holographic RIS observations through an equivalent measurement matrix while keeping the analysis in the discrete symbol domain.

B. Far-Field Channel Model

As shown in Fig. 1, we consider a holographic RIS receiver composed of a uniform planar array placed on the yz-plane. The array consists of N_y elements along the y-axis and N_z elements along the z-axis, resulting in a total of

$$N_{\mathrm{r}} = N_y N_z \tag{18}$$

receiving elements. The position vector of the (n, m)-th RIS element is denoted by

$$\mathbf{r}_{n,m} = [0, y_n, z_m]^T, \quad n = 1, \ldots, N_y, \quad m = 1, \ldots, N_z, \tag{19}$$

where

$$y_n = \left(n - \frac{N_y + 1}{2}\right)d_y, \quad z_m = \left(m - \frac{N_z + 1}{2}\right)d_z. \tag{20}$$

Here, d_y and d_z denote the inter-element spacings along the y- and z-axes, respectively. The centered coordinate system is adopted for notational convenience; using a corner element as the origin leads only to a user-dependent common phase shift, which can be absorbed into the complex channel gain.

Under the far-field assumption, the impinging signal from each user can be modeled as a plane wave across the RIS aperture. Let θ_l and ϕ_l denote the angle of arrival (AoA) of the l-th user, where θ_l is the polar angle with respect to the array broadside, i.e., the positive x-axis, and ϕ_l is the azimuth angle in the yz-plane measured from the positive y-axis. The corresponding wave vector is given by

$$\mathbf{k}_l = [k_{l,x}, k_{l,y}, k_{l,z}]^T, \tag{21}$$

with

$$k_{l,x} = k_0 \cos\theta_l, \quad k_{l,y} = k_0 \sin\theta_l \cos\phi_l, \quad k_{l,z} = k_0 \sin\theta_l \sin\phi_l, \tag{22}$$

where $k_0 = \frac{2\pi}{\lambda}$ is the free-space wavenumber and λ is the carrier wavelength.

Following the plane-wave model, the phase of the incident field from user l at the (n,m)-th RIS element is determined by the projection of the wave vector onto the element position, i.e.,

$$\psi_{n,m}^{(l)} = \mathbf{k}_l^T \mathbf{r}_{n,m} = k_{l,y} y_n + k_{l,z} z_m. \tag{23}$$

Since the RIS aperture is assumed to be sufficiently small compared with the propagation distance, the amplitude variation of the incident wave across different RIS elements is neglected for each user. Therefore, the spatial variation across the RIS elements is mainly characterized by the phase profile in (24), while the large-scale path loss, shadowing, and user-dependent phase offset are incorporated into a complex channel gain.

Accordingly, the channel vector from the l-th user to all RIS receiving elements is modeled as

$$\mathbf{h}_l = \alpha_l \mathbf{v}_l \in \mathbb{C}^{N_\mathrm{r} \times 1}, \tag{24}$$

where $\alpha_l \in \mathbb{C}$ denotes the complex channel gain of user l, and $\mathbf{v}_l$ is the normalized array steering vector. The element of $\mathbf{v}_l$ corresponding to the (n,m)-th RIS element is given by

$$[\mathbf{v}_l]_{\iota(n,m)} = e^{j(k_{l,y} y_n + k_{l,z} z_m)}, \tag{25}$$

where

$$\iota(n,m) = (n-1)N_z + m \tag{26}$$

maps the two-dimensional element index (n,m) to the one-dimensional vector index.

Equivalently, the array steering vector can be written as the Kronecker product of two one-dimensional steering vectors:

$$\mathbf{v}_l = \mathbf{a}_y(\theta_l, \phi_l) \otimes \mathbf{a}_z(\theta_l, \phi_l), \tag{27}$$

where $\otimes$ denotes the Kronecker product,

$$\mathbf{a}_y(\theta_l, \phi_l) = \left[e^{jk_{l,y} y_1}, e^{jk_{l,y} y_2}, \ldots, e^{jk_{l,y} y_{N_y}}\right]^T \in \mathbb{C}^{N_y \times 1}, \tag{28}$$

and

$$\mathbf{a}_z(\theta_l, \phi_l) = \left[e^{jk_{l,z} z_1}, e^{jk_{l,z} z_2}, \ldots, e^{jk_{l,z} z_{N_z}}\right]^T \in \mathbb{C}^{N_z \times 1}. \tag{29}$$

Collecting the channel vectors of all L users, the multi-user channel matrix between the users and the RIS receiver is expressed as

$$\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_L] \in \mathbb{C}^{N_\mathrm{r} \times L}. \tag{30}$$

Using (25), it can be decomposed as

$$\mathbf{H} = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_L] \operatorname{diag}(\alpha_1, \alpha_2, \ldots, \alpha_L). \tag{31}$$

For compactness, define

$$\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_L] \in \mathbb{C}^{N_\mathrm{r} \times L} \tag{32}$$

as the RIS array manifold matrix, and

$$\boldsymbol{\Gamma} = \operatorname{diag}(\alpha_1, \alpha_2, \ldots, \alpha_L) \in \mathbb{C}^{L \times L} \tag{33}$$

as the diagonal matrix of complex channel gains. Then the multi-user channel matrix is compactly written as

$$\mathbf{H} = \mathbf{V}\boldsymbol{\Gamma}. \tag{34}$$

This decomposition separates the directional spatial response of the RIS, represented by $\mathbf{V}$, from the user-dependent complex propagation gains, represented by $\boldsymbol{\Gamma}$. It will be used in the subsequent receiver model to characterize how the pulse-shaped user signals are spatially mapped onto the holographic RIS elements.

C. Reference-Assisted Holographic Reception Model

At the receiver side, the holographic RIS collects the superimposed signals from all users through its N_r receiving elements. Using the multi-user channel matrix $\mathbf{H} \in \mathbb{C}^{N_\mathrm{r} \times L}$ defined in the previous subsection, the complex baseband signal impinging on the RIS at time t is written as

$$\mathbf{r}(t) = \mathbf{H}\mathbf{x}(t), \tag{35}$$

where

$$\mathbf{x}(t) = [x_1(t), x_2(t), \ldots, x_L(t)]^T \in \mathbb{C}^L \tag{36}$$

is the transmitted complex baseband waveform vector of all users.

To enable phase information recovery from intensity-only observations, a known reference wave is introduced at the RIS receiver. Let

$$\mathbf{b} = [b_1, b_2, \ldots, b_{N_\mathrm{r}}]^T \in \mathbb{C}^{N_\mathrm{r}} \tag{37}$$

denote the equivalent complex baseband reference vector, where

$$b_i = \rho_i e^{j\varphi_i}, \quad i = 1, 2, \ldots, N_\mathrm{r}. \tag{38}$$

Here, ρ_i and φ_i are the known amplitude and phase of the reference wave injected at the i-th RIS element, respectively. The corresponding passband reference signal is

$$b_i^\mathrm{RF}(t) = \sqrt{2}\Re\left\{b_i e^{j2\pi f_c t}\right\} = \sqrt{2}\rho_i \cos(2\pi f_c t + \varphi_i). \tag{39}$$

After superimposing the received signal and the reference wave, the equivalent complex baseband field at the RIS is given by

$$\mathbf{u}(t) = \mathbf{r}(t) + \mathbf{b} = \mathbf{H}\mathbf{x}(t) + \mathbf{b}. \tag{40}$$

Equivalently, the real-valued RF signal at the i-th RIS element can be expressed as

$$u_i^\mathrm{RF}(t) = \sqrt{2}\Re\left\{u_i(t) e^{j2\pi f_c t}\right\} + \nu_i^\mathrm{RF}(t), \tag{41}$$

where $u_i(t)$ is the i-th element of $\mathbf{u}(t)$ and $\nu_i^\mathrm{RF}(t)$ denotes the RF noise.

The holographic RIS receiver is assumed to employ square-law detection at each element. After low-pass filtering or time averaging over a detection interval T_0, the intensity measurement at the i-th element is modeled as

$$z_i(t) = \frac{1}{T_0} \int_t^{t+T_0} \left| u_i^{\mathrm{RF}}(\tau) \right|^2 d\tau. \quad (42)$$

Under the narrowband and slowly varying envelope assumptions within the detection interval, the high-frequency components are removed by the low-pass operation, yielding the equivalent baseband intensity observation

$$z_i(t) = |[\mathbf{H}\mathbf{x}(t)]_i + b_i|^2 + w_i(t), \quad i = 1, 2, \ldots, N_{\mathrm{r}}, \quad (43)$$

where $w_i(t)$ denotes the effective post-detection noise and modeling error.

Stacking the observations of all RIS elements gives

$$\mathbf{z}(t) = |\mathbf{H}\mathbf{x}(t) + \mathbf{b}|^2 + \mathbf{w}(t), \quad (44)$$

where

$$\mathbf{z}(t) = [z_1(t), z_2(t), \ldots, z_{N_{\mathrm{r}}}(t)]^T \in \mathbb{R}^{N_{\mathrm{r}}}, \quad (45)$$

$\mathbf{w}(t) \in \mathbb{R}^{N_{\mathrm{r}}}$ is the effective intensity-domain noise vector, and the squared modulus in (45) is applied element-wise.

For subsequent symbol-level analysis, we evaluate (45) at the discrete observation instants $\{t_n\}_{n=0}^{N_t-1}$. Let

$$\mathbf{x}[n] \triangleq \mathbf{x}(t_n), \quad \mathbf{z}[n] \triangleq \mathbf{z}(t_n), \quad \mathbf{w}[n] \triangleq \mathbf{w}(t_n). \quad (46)$$

Then the discrete-time reference-assisted intensity measurement is

$$\mathbf{z}[n] = |\mathbf{H}\mathbf{x}[n] + \mathbf{b}|^2 + \mathbf{w}[n], \quad n = 0, 1, \ldots, N_t - 1. \quad (47)$$

Finally, using the transmitter-side equivalent matrix model in (18), the complete block-level observation can be compactly represented as

$$\mathbf{z} = |\mathbf{G}\mathbf{s} + \mathbf{b}_{\mathrm{eq}}|^2 + \mathbf{w}, \quad (48)$$

where $\mathbf{s} \in \mathcal{S}^{LK}$ is the stacked transmitted symbol vector, $\mathbf{G}$ is the equivalent measurement matrix incorporating pulse shaping, propagation, and RIS spatial responses, $\mathbf{b}_{\mathrm{eq}}$ is the stacked equivalent reference vector, and $\mathbf{z}$ and $\mathbf{w}$ collect all intensity observations and post-detection noise samples, respectively. The compact model in (49) will serve as the basis for the subsequent identifiability analysis and detector design.

D. Two-Phase Transmission Protocol and Unified Observation Model

Based on the reference-assisted holographic reception model, we consider a two-phase transmission protocol consisting of a training phase and a data transmission phase. In the training phase, the users transmit known pilot waveforms, and the RIS receiver estimates the channel from intensity-only measurements. In the data transmission phase, the users transmit information-bearing symbols, and the receiver detects the transmitted symbols based on the estimated channel. Both phases follow the same reference-assisted intensity observation principle, but the unknown variables are different.

1) Training Phase: During the training phase, all users transmit known pilot signals. Let $\mathbf{x}_{\mathrm{p}}[t] \in \mathbb{C}^L$ denote the known multi-user pilot vector at the t-th pilot observation, where $t = 0, 1, \ldots, T_{\mathrm{p}}-1$ and T_{p} is the number of pilot observations. The corresponding reference vector is denoted by $\mathbf{b}_{\mathrm{p}}[t] \in \mathbb{C}^{N_{\mathrm{r}}}$. According to (48), the received intensity measurement in the training phase is given by

$$\mathbf{z}_{\mathrm{p}}[t] = |\mathbf{H}\mathbf{x}_{\mathrm{p}}[t] + \mathbf{b}_{\mathrm{p}}[t]|^2 + \mathbf{w}_{\mathrm{p}}[t], \quad t = 0, 1, \ldots, T_{\mathrm{p}} - 1, \quad (49)$$

where $\mathbf{z}_{\mathrm{p}}[t] \in \mathbb{R}^{N_{\mathrm{r}}}$ is the intensity observation vector, and $\mathbf{w}_{\mathrm{p}}[t] \in \mathbb{R}^{N_{\mathrm{r}}}$ denotes the effective post-detection noise.

For compact representation, define the vectorized channel as

$$\mathbf{h} = \mathrm{vec}(\mathbf{H}) \in \mathbb{C}^{N_{\mathrm{r}}L}. \quad (50)$$

Using the identity

$$\mathbf{H}\mathbf{x}_{\mathrm{p}}[t] = \left(\mathbf{x}_{\mathrm{p}}^T[t] \otimes \mathbf{I}_{N_{\mathrm{r}}}\right) \mathbf{h}, \quad (51)$$

(50) can be rewritten as

$$\mathbf{z}_{\mathrm{p}}[t] = |\mathbf{G}_{\mathrm{p}}[t]\mathbf{h} + \mathbf{b}_{\mathrm{p}}[t]|^2 + \mathbf{w}_{\mathrm{p}}[t], \quad (52)$$

where

$$\mathbf{G}_{\mathrm{p}}[t] = \mathbf{x}_{\mathrm{p}}^T[t] \otimes \mathbf{I}_{N_{\mathrm{r}}} \in \mathbb{C}^{N_{\mathrm{r}} \times N_{\mathrm{r}}L}. \quad (53)$$

Stacking all pilot observations yields the block-level training model

$$\mathbf{z}_{\mathrm{p}} = |\mathbf{G}_{\mathrm{p}}\mathbf{h} + \mathbf{b}_{\mathrm{p}}|^2 + \mathbf{w}_{\mathrm{p}}, \quad (54)$$

where $\mathbf{z}_{\mathrm{p}}$, $\mathbf{b}_{\mathrm{p}}$, and $\mathbf{w}_{\mathrm{p}}$ collect the corresponding quantities over all pilot observations, and

$$\mathbf{G}_{\mathrm{p}} = \begin{bmatrix} \mathbf{G}_{\mathrm{p}}[0] \\ \mathbf{G}_{\mathrm{p}}[1] \\ \vdots \\ \mathbf{G}_{\mathrm{p}}[T_{\mathrm{p}} - 1] \end{bmatrix}. \quad (55)$$

If the far-field channel structure in (35) is exploited, the unknown channel vector $\mathbf{h}$ can be further parameterized by a lower-dimensional channel parameter vector, such as the complex path-gain vector when the angles of arrival are known or pre-estimated.

2) Data Transmission Phase: After the training phase, the receiver obtains an estimate of the channel, denoted by $\hat{\mathbf{H}}$. In the data transmission phase, the users transmit unknown information-bearing symbols. Let

$$\mathbf{s}_{\mathrm{d}} \in \mathcal{S}^{LK_{\mathrm{d}}} \quad (56)$$

denote the stacked data-symbol vector within one data block, where K_{d} is the number of data symbols per user. Following the transmitter-side equivalent model, the sampled data waveform can be expressed as

$$\tilde{\mathbf{x}}_{\mathrm{d}} = \mathbf{P}_{\mathrm{tx},d}\mathbf{s}_{\mathrm{d}} + \mathbf{x}_{0,d}, \quad (57)$$

where $\mathbf{P}_{\mathrm{tx},d}$ is the data-phase pulse-shaping matrix and $\mathbf{x}_{0,d}$ is the deterministic transmit-side pilot or bias component, if present.

Using the estimated channel $\hat{\mathbf{H}}$, the block-level data-phase intensity observation can be written as

$$\mathbf{z}_{\mathrm{d}} = \left|\mathbf{G}_{\mathrm{d}}(\hat{\mathbf{H}})\mathbf{s}_{\mathrm{d}} + \mathbf{b}_{\mathrm{d,eq}}\right|^2 + \mathbf{w}_{\mathrm{d}}, \quad (58)$$

where $\mathbf{G}_{\mathrm{d}}(\hat{\mathbf{H}})$ is the equivalent data-phase measurement matrix determined by the estimated channel, the pulse-shaping matrix, and the RIS observation structure. The vector $\mathbf{b}_{\mathrm{d,eq}}$ denotes the equivalent known offset in the data phase, which includes the receiver-side reference wave and the contribution of any deterministic transmit-side bias. The vector $\mathbf{w}_{\mathrm{d}}$ represents the effective post-detection noise.

Equations (55) and (59) show that both channel estimation and data detection can be written in the common reference-assisted intensity measurement form

$$\mathbf{z} = \left|\mathbf{G}\boldsymbol{\theta} + \mathbf{b}_{\mathrm{eq}}\right|^2 + \mathbf{w}. \tag{59}$$

In the training phase, the unknown variable is the channel-related parameter, e.g., $\boldsymbol{\theta} = \mathbf{h}$ or a lower-dimensional parameterization of $\mathbf{H}$. In the data transmission phase, the unknown variable is the finite-alphabet symbol vector, i.e., $\boldsymbol{\theta} = \mathbf{s}_{\mathrm{d}}$. Therefore, the two phases share the same phaseless observation structure, while their identifiability conditions and recovery algorithms differ due to the different structures of the unknown variables. The unified model in (60) will serve as the basis for the subsequent identifiability analysis and detector design.

III. Structural Analysis of Reference-Assisted Energy Detection

In this section, we investigate the identifiability of the proposed reference-assisted holographic reception model. As shown in the previous section, both the pilot-based channel estimation phase and the data transmission phase lead to intensity-only observations with a known reference wave. Although the unknown variables in the two phases are different, they share the same mathematical structure. This motivates a unified reference-assisted phase retrieval formulation, which serves as the basis for the subsequent identifiability analysis.

A. Unified Reference-Assisted Phase Retrieval Model

From the two-phase transmission protocol in Section II, the training-phase and data-phase observations can be written in the common form

$$\mathbf{z} = \left|\mathbf{G}\boldsymbol{\theta} + \mathbf{b}\right|^2 + \mathbf{w}, \tag{60}$$

where $\mathbf{z} \in \mathbb{R}^Q$ denotes the collected intensity measurements, $\mathbf{G} \in \mathbb{C}^{Q\times D}$ is the equivalent measurement matrix, $\boldsymbol{\theta} \in \mathbb{C}^D$ is the unknown parameter vector, $\mathbf{b} \in \mathbb{C}^Q$ is the known equivalent reference vector, and $\mathbf{w} \in \mathbb{R}^Q$ represents the effective post-detection noise. The squared modulus in (61) is applied element-wise, and Q denotes the total number of intensity observations.

The physical meaning of $\boldsymbol{\theta}$ and $\mathbf{G}$ depends on the considered transmission phase. In the training phase, the unknown variable is the channel-related parameter. For example, when the full channel matrix is estimated, one has

$$\boldsymbol{\theta} = \mathbf{h} = \mathrm{vec}(\mathbf{H}), \tag{61}$$

and the corresponding measurement matrix is given by the stacked pilot-dependent matrix $\mathbf{G}_{\mathrm{p}}$ in (55). When the far-field structure $\mathbf{H} = \mathbf{V}\boldsymbol{\Gamma}$ is exploited, $\boldsymbol{\theta}$ can be further reduced to a lower-dimensional channel parameter vector, such as the complex path-gain vector. In the data transmission phase, the channel is assumed to be known or estimated from the training phase, and the unknown variable becomes the finite-alphabet data-symbol vector, namely

$$\boldsymbol{\theta} = \mathbf{s}_{\mathrm{d}} \in \mathcal{S}^{LK_{\mathrm{d}}}, \tag{62}$$

with the corresponding equivalent measurement matrix $\mathbf{G}_{\mathrm{d}}(\hat{\mathbf{H}})$ defined in (59).

For notational compactness, define the noiseless reference-assisted intensity mapping as

$$\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) \triangleq \left|\mathbf{G}\boldsymbol{\theta} + \mathbf{b}\right|^2. \tag{63}$$

Then (61) can be written as

$$\mathbf{z} = \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) + \mathbf{w}. \tag{64}$$

The corresponding recovery problem is to infer $\boldsymbol{\theta}$ from the real-valued intensity observation $\mathbf{z}$, given $\mathbf{G}$ and $\mathbf{b}$. A general least-squares formulation is

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}\in\mathcal{X}} \left\|\mathbf{z} - \left|\mathbf{G}\boldsymbol{\theta} + \mathbf{b}\right|^2\right\|_2^2, \tag{65}$$

where $\mathcal{X}$ denotes the feasible set of the unknown variable. For channel estimation, $\mathcal{X}$ is typically a continuous parameter space determined by the adopted channel model. For data detection, $\mathcal{X}$ is a finite constellation set, e.g.,

$$\mathcal{X} = \mathcal{S}^{LK_{\mathrm{d}}}. \tag{66}$$

The key feature of (61) is the presence of the known reference vector $\mathbf{b}$. Expanding the noiseless observation gives

$$\left|\mathbf{G}\boldsymbol{\theta} + \mathbf{b}\right|^2 = \left|\mathbf{G}\boldsymbol{\theta}\right|^2 + \left|\mathbf{b}\right|^2 + 2\,\mathrm{Re}\left\{\mathbf{b}^* \odot \mathbf{G}\boldsymbol{\theta}\right\}, \tag{67}$$

where $\odot$ denotes the Hadamard product. Compared with conventional phaseless measurements of the form $|\mathbf{G}\boldsymbol{\theta}|^2$, the reference-assisted model introduces the interference term

$$2\,\mathrm{Re}\left\{\mathbf{b}^* \odot \mathbf{G}\boldsymbol{\theta}\right\}, \tag{68}$$

which depends on the phase of $\mathbf{G}\boldsymbol{\theta}$ relative to the known reference wave. Therefore, the reference wave acts as a phase anchor that converts part of the hidden phase information into observable intensity variations.

The subsequent analysis focuses on the following question: under what conditions does the mapping $\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta})$ uniquely determine the unknown variable $\boldsymbol{\theta}$ within its feasible set $\mathcal{X}$? For the training phase, this question corresponds to the identifiability of channel parameters from reference-assisted intensity measurements. For the data transmission phase, it corresponds to the uniqueness of finite-alphabet symbol detection after square-law observation. Although both problems follow the same observation model, their identifiability conditions differ because the channel parameters are generally continuous, whereas the transmitted symbols belong to a discrete constellation set.

B. Ambiguity Removal by the Reference Wave

We first discuss the role of the reference wave in removing the inherent ambiguity of intensity-only measurements. For conventional phaseless observations without a reference wave, the noiseless measurement is given by

$$\mathbf{z}_0 = |\mathbf{G}\boldsymbol{\theta}|^2. \tag{69}$$

This model suffers from a fundamental global phase ambiguity. Specifically, for any phase rotation $e^{j\varphi}$ with $\varphi \in [0, 2\pi)$, we have

$$|\mathbf{G}(e^{j\varphi}\boldsymbol{\theta})|^2 = |e^{j\varphi}\mathbf{G}\boldsymbol{\theta}|^2 = |\mathbf{G}\boldsymbol{\theta}|^2. \tag{70}$$

Therefore, $\boldsymbol{\theta}$ and $e^{j\varphi}\boldsymbol{\theta}$ cannot be distinguished from (70). This ambiguity is particularly problematic for communication systems, where the absolute phase of the recovered channel or data symbols directly affects coherent combining and constellation decision.

In contrast, the proposed reference-assisted observation model is given by

$$\mathbf{z} = |\mathbf{G}\boldsymbol{\theta} + \mathbf{b}|^2, \tag{71}$$

where $\mathbf{b}$ is a known reference vector. Let

$$\mathbf{u} = \mathbf{G}\boldsymbol{\theta} \tag{72}$$

denote the effective field generated by the unknown parameter vector. Then the q-th noiseless intensity observation can be written as

$$z_q = |u_q + b_q|^2 = |u_q|^2 + |b_q|^2 + 2\,\mathrm{Re}\{b_q^* u_q\}. \tag{73}$$

The first term $|u_q|^2$ is the conventional phaseless component, while the third term

$$2\,\mathrm{Re}\{b_q^* u_q\} \tag{74}$$

is the reference-induced interference component. This term depends on the relative phase between the unknown field u_q and the known reference wave b_q, and therefore provides a phase anchor for the intensity measurement.

To see how the reference wave removes the global phase ambiguity, consider the globally rotated parameter vector $e^{j\varphi}\boldsymbol{\theta}$. The corresponding noiseless observation is

$$|\mathbf{G}(e^{j\varphi}\boldsymbol{\theta}) + \mathbf{b}|^2 = |e^{j\varphi}\mathbf{u} + \mathbf{b}|^2. \tag{75}$$

The difference between the rotated and original intensity observations is

$$\begin{aligned} &|e^{j\varphi}\mathbf{u} + \mathbf{b}|^2 - |\mathbf{u} + \mathbf{b}|^2 \\ &= 2\,\mathrm{Re}\left\{\mathbf{b}^* \odot \mathbf{u}\left(e^{j\varphi} - 1\right)\right\}, \end{aligned} \tag{76}$$

where the operation is applied element-wise. Hence, the rotated vector $e^{j\varphi}\boldsymbol{\theta}$ produces the same reference-assisted intensity measurement as $\boldsymbol{\theta}$ if and only if

$$\mathrm{Re}\left\{b_q^* u_q \left(e^{j\varphi} - 1\right)\right\} = 0, \quad q = 1, 2, \ldots, Q. \tag{77}$$

Equation (78) shows that the global phase ambiguity is no longer automatically preserved when a nonzero reference wave is present. Let

$$b_q^* u_q = \rho_q e^{j\beta_q}, \quad \rho_q \geq 0, \tag{78}$$

for all observation indices satisfying $\rho_q > 0$. Then (78) is equivalent to

$$\cos(\beta_q + \varphi) = \cos(\beta_q), \quad \forall q \text{ with } \rho_q > 0. \tag{79}$$

For a nontrivial phase rotation $\varphi \not\equiv 0 \pmod{2\pi}$ to remain ambiguous, the same phase rotation must satisfy (80) for all active observations. This can only occur under special phase alignments between the effective field $\mathbf{u}$ and the reference vector $\mathbf{b}$. In particular, if there exist two active observation indices q_1 and q_2 such that

$$\beta_{q_1} \not\equiv \beta_{q_2} \pmod{\pi}, \tag{80}$$

then no nontrivial global phase rotation can produce the same reference-assisted intensity measurement. Thus, under the nondegenerate condition in (81), the global phase ambiguity in (71) is removed by the reference wave.

The above discussion reveals two important implications. First, the reference wave does not merely increase the received power; instead, it introduces a deterministic interference term that converts relative phase information into measurable intensity variations. Second, ambiguity removal depends on the joint structure of the reference vector $\mathbf{b}$ and the effective field $\mathbf{G}\boldsymbol{\theta}$. If the reference phases are poorly designed, or if the effective measurements $\mathbf{G}\boldsymbol{\theta}$ are aligned with the reference wave in a degenerate manner, residual ambiguities may still exist. This motivates the design of reference phase diversity across different RIS elements, time slots, or measurement configurations.

More generally, for two distinct feasible parameters $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X}$, they are indistinguishable under the noiseless reference-assisted model if

$$|\mathbf{G}\boldsymbol{\theta}_1 + \mathbf{b}|^2 = |\mathbf{G}\boldsymbol{\theta}_2 + \mathbf{b}|^2. \tag{81}$$

By expanding both sides of (82), this condition can be equivalently written as

$$\mathrm{Re}\left\{[\mathbf{G}(\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2) + 2\mathbf{b}]^* \odot \mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)\right\} = \mathbf{0}. \tag{82}$$

Equation (83) will be used in the following subsections to characterize the identifiability of channel estimation and finite-alphabet data detection. It shows that reference-assisted identifiability is determined jointly by the effective measurement matrix $\mathbf{G}$, the feasible set $\mathcal{X}$, and the reference vector $\mathbf{b}$.

C. Identifiability for Channel Estimation and Symbol Detection

Based on the unified model in (61), this subsection analyzes the identifiability of the unknown parameter vector $\boldsymbol{\theta}$ from reference-assisted intensity measurements. The key question is whether two different feasible parameters can generate the same noiseless intensity observation, i.e.,

$$\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_1) = \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_2), \quad \boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2. \tag{83}$$

As shown in (83), two feasible parameters are indistinguishable if and only if

$$\mathrm{Re}\left\{[\mathbf{G}(\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2) + 2\mathbf{b}]^* \odot \mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)\right\} = \mathbf{0}. \tag{84}$$

This condition reveals the complementary roles of the effective measurement matrix $\mathbf{G}$ and the reference vector $\mathbf{b}$. The matrix $\mathbf{G}$ determines whether the difference between two feasible parameters can be observed in the complex field domain, whereas the reference vector $\mathbf{b}$ determines whether such a complex-field difference can be projected into the real-valued intensity domain.

1) Necessary Condition on the Effective Measurement Matrix: Let the feasible set of the unknown parameter be denoted by $\mathcal{X}$, and define its difference set as

$$\mathcal{D}_{\mathcal{X}} = \{\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2 : \boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X}, \boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2\}. \tag{85}$$

A necessary condition for identifiability over $\mathcal{X}$ is

$$\mathrm{Null}(\mathbf{G}) \cap \mathcal{D}_{\mathcal{X}} = \varnothing. \tag{86}$$

Indeed, if there exists a nonzero difference vector $\Delta\boldsymbol{\theta} \in \mathcal{D}_{\mathcal{X}}$ such that

$$\mathbf{G}\Delta\boldsymbol{\theta} = \mathbf{0}, \tag{87}$$

then two different feasible parameters $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$ satisfying $\Delta\boldsymbol{\theta} = \boldsymbol{\theta}_1 - \boldsymbol{\theta}_2$ produce the same complex field,

$$\mathbf{G}\boldsymbol{\theta}_1 = \mathbf{G}\boldsymbol{\theta}_2. \tag{88}$$

Consequently, for any reference vector $\mathbf{b}$, one has

$$|\mathbf{G}\boldsymbol{\theta}_1 + \mathbf{b}|^2 = |\mathbf{G}\boldsymbol{\theta}_2 + \mathbf{b}|^2. \tag{89}$$

Therefore, the reference wave cannot recover information that has already been lost in the null space of $\mathbf{G}$.

For continuous parameter spaces with nonempty interior, (87) usually requires $\mathbf{G}$ to be full column rank, i.e.,

$$\mathrm{rank}(\mathbf{G}) = D, \tag{90}$$

where D is the dimension of $\boldsymbol{\theta}$. For finite-alphabet data detection, however, the requirement is weaker: $\mathbf{G}$ does not need to be full column rank over the entire continuous space, but it must not annihilate any valid constellation difference in $\mathcal{D}_{\mathcal{X}}$.

The stability of this condition can be characterized by the minimum field-domain separation

$$d_{\min}^{(F)} = \min_{\Delta\boldsymbol{\theta} \in \mathcal{D}_{\mathcal{X}}} \|\mathbf{G}\Delta\boldsymbol{\theta}\|_2. \tag{91}$$

For a continuous feasible set, the corresponding local sensitivity is governed by the smallest singular value $\sigma_{\min}(\mathbf{G})$. If $\mathbf{G}$ is full column rank, then

$$\|\mathbf{G}\Delta\boldsymbol{\theta}\|_2 \geq \sigma_{\min}(\mathbf{G})\|\Delta\boldsymbol{\theta}\|_2. \tag{92}$$

Hence, a larger $\sigma_{\min}(\mathbf{G})$ or a smaller condition number

$$\kappa(\mathbf{G}) = \frac{\sigma_{\max}(\mathbf{G})}{\sigma_{\min}(\mathbf{G})} \tag{93}$$

implies a more stable measurement mapping.

2) Local Identifiability for Channel Estimation: In the training phase, the unknown variable is generally a continuous channel-related parameter. For the full channel estimation model in (55), the unknown is

$$\boldsymbol{\theta} = \mathbf{h} = \mathrm{vec}(\mathbf{H}). \tag{94}$$

Since the channel is continuous, global identifiability is difficult to characterize in closed form. A useful and tractable criterion is local identifiability, which can be studied through the real-valued Jacobian of the intensity mapping.

Let

$$\mathbf{u} = \mathbf{G}\boldsymbol{\theta} + \mathbf{b}. \tag{95}$$

For a small perturbation $d\boldsymbol{\theta}$, the first-order variation of the noiseless intensity observation is

$$d\mathbf{z} = 2\,\mathrm{Re}\left\{\mathrm{diag}(\mathbf{u}^*)\mathbf{G}d\boldsymbol{\theta}\right\}. \tag{96}$$

Define the real-valued parameter vector as

$$\bar{\boldsymbol{\theta}} = \begin{bmatrix}\mathrm{Re}\{\boldsymbol{\theta}\}^T & \mathrm{Im}\{\boldsymbol{\theta}\}^T\end{bmatrix}^T \in \mathbb{R}^{2D}. \tag{97}$$

Then the real-valued Jacobian of the mapping $\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta})$ is

$$\mathbf{J}(\boldsymbol{\theta}) = 2\left[\mathrm{Re}\left\{\mathrm{diag}(\mathbf{u}^*)\mathbf{G}\right\}, -\mathrm{Im}\left\{\mathrm{diag}(\mathbf{u}^*)\mathbf{G}\right\}\right] \in \mathbb{R}^{Q\times 2D}. \tag{98}$$

The channel parameter $\boldsymbol{\theta}$ is locally identifiable if

$$\mathrm{rank}\left(\mathbf{J}(\boldsymbol{\theta})\right) = 2D. \tag{99}$$

This condition also implies that the number of intensity observations must satisfy

$$Q \geq 2D. \tag{100}$$

The conditioning of the Jacobian further determines the stability of channel estimation. When $\mathbf{J}(\boldsymbol{\theta})$ is full column rank, small intensity perturbations lead to bounded parameter perturbations, and the sensitivity is controlled by

$$\sigma_{\min}\left(\mathbf{J}(\boldsymbol{\theta})\right). \tag{101}$$

Therefore, pilot and reference design should aim to increase $\sigma_{\min}(\mathbf{J}(\boldsymbol{\theta}))$ or reduce

$$\kappa\left(\mathbf{J}(\boldsymbol{\theta})\right) = \frac{\sigma_{\max}(\mathbf{J}(\boldsymbol{\theta}))}{\sigma_{\min}(\mathbf{J}(\boldsymbol{\theta}))}. \tag{102}$$

For the full channel vectorization model, the training measurement matrix is

$$\mathbf{G}_{\mathrm{p}} = \mathbf{X}_{\mathrm{p}} \otimes \mathbf{I}_{N_{\mathrm{r}}}, \tag{103}$$

where

$$\mathbf{X}_{\mathrm{p}} = \begin{bmatrix}\mathbf{x}_{\mathrm{p}}^T[0] \\ \mathbf{x}_{\mathrm{p}}^T[1] \\ \vdots \\ \mathbf{x}_{\mathrm{p}}^T[T_{\mathrm{p}}-1]\end{bmatrix} \in \mathbb{C}^{T_{\mathrm{p}}\times L} \tag{104}$$

is the pilot matrix. Thus,

$$\mathrm{rank}(\mathbf{G}_{\mathrm{p}}) = N_{\mathrm{r}}\,\mathrm{rank}(\mathbf{X}_{\mathrm{p}}). \tag{105}$$

A necessary condition for full-channel estimation is

$$\mathrm{rank}(\mathbf{X}_{\mathrm{p}}) = L, \quad T_{\mathrm{p}} \geq L. \tag{106}$$

Moreover, the condition number satisfies

$$\kappa(\mathbf{G}_{\mathrm{p}}) = \kappa(\mathbf{X}_{\mathrm{p}}), \tag{107}$$

which indicates that orthogonal or well-conditioned pilot sequences are beneficial for stable channel recovery.

If the far-field structure $\mathbf{H} = \mathbf{V}\boldsymbol{\Gamma}$ is exploited and the angles of arrival are known or pre-estimated, the unknown channel parameter can be reduced to the complex gain vector

$$\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \ldots, \alpha_L]^T. \tag{108}$$

At the t-th pilot observation, the received field can be written as

$$\mathbf{H}\mathbf{x}_{\mathrm{p}}[t] = \mathbf{V}\,\mathrm{diag}(\mathbf{x}_{\mathrm{p}}[t])\boldsymbol{\alpha}. \tag{109}$$

Thus, the corresponding gain-domain measurement matrix is

$$\mathbf{G}_{\alpha}[t] = \mathbf{V}\,\mathrm{diag}(\mathbf{x}_{\mathrm{p}}[t]). \tag{110}$$

Stacking all pilot observations gives

$$\mathbf{G}_{\alpha} = \begin{bmatrix} \mathbf{V}\,\mathrm{diag}(\mathbf{x}_{\mathrm{p}}[0]) \\ \mathbf{V}\,\mathrm{diag}(\mathbf{x}_{\mathrm{p}}[1]) \\ \vdots \\ \mathbf{V}\,\mathrm{diag}(\mathbf{x}_{\mathrm{p}}[T_{\mathrm{p}}-1]) \end{bmatrix}. \tag{111}$$

In this structured case, the identifiability of $\boldsymbol{\alpha}$ depends jointly on the array manifold $\mathbf{V}$ and the pilot matrix. In particular, if $\mathbf{V}$ is full column rank and each user is sufficiently excited by the pilot sequence, then $\mathbf{G}_{\alpha}$ can be full column rank. The conditioning of $\mathbf{V}$ is determined by the angular separation of the users; closely spaced angles lead to highly correlated steering vectors, a smaller $\sigma_{\min}(\mathbf{V})$, and a more ill-conditioned channel estimation problem.

3) Discrete Identifiability for Data Symbol Detection: In the data transmission phase, the unknown variable is the finite-alphabet symbol vector

$$\boldsymbol{\theta} = \mathbf{s}_{\mathrm{d}} \in \mathcal{S}^{LK_{\mathrm{d}}}. \tag{112}$$

Unlike channel estimation, the feasible set is finite. Therefore, symbol identifiability can be characterized globally over the constellation set.

Define the constellation difference set as

$$\mathcal{D}_{\mathrm{s}} = \left\{\mathbf{s}_1 - \mathbf{s}_2 : \mathbf{s}_1, \mathbf{s}_2 \in \mathcal{S}^{LK_{\mathrm{d}}}, \mathbf{s}_1 \neq \mathbf{s}_2\right\}. \tag{113}$$

A necessary condition for symbol identifiability is

$$\mathrm{Null}(\mathbf{G}_{\mathrm{d}}) \cap \mathcal{D}_{\mathrm{s}} = \varnothing. \tag{114}$$

If this condition is violated, two different symbol vectors produce the same complex field and cannot be separated by any reference wave or detector.

When (115) holds, the remaining ambiguity is caused by the square-law operation. The data symbols are identifiable under the reference-assisted intensity model if and only if

$$|\mathbf{G}_{\mathrm{d}}\mathbf{s}_1 + \mathbf{b}_{\mathrm{d}}|^2 \neq |\mathbf{G}_{\mathrm{d}}\mathbf{s}_2 + \mathbf{b}_{\mathrm{d}}|^2, \quad \forall \mathbf{s}_1 \neq \mathbf{s}_2. \tag{115}$$

Equivalently, using (85), every nonzero symbol pair must satisfy

$$\mathrm{Re}\left\{[\mathbf{G}_{\mathrm{d}}(\mathbf{s}_1 + \mathbf{s}_2) + 2\mathbf{b}_{\mathrm{d}}]^* \odot \mathbf{G}_{\mathrm{d}}(\mathbf{s}_1 - \mathbf{s}_2)\right\} \neq \mathbf{0}. \tag{116}$$

The stability of finite-alphabet detection can be quantified by the energy-domain minimum distance

$$d_{\min}^{(E)} = \min_{\mathbf{s}_1 \neq \mathbf{s}_2} \left\| |\mathbf{G}_{\mathrm{d}}\mathbf{s}_1 + \mathbf{b}_{\mathrm{d}}|^2 - |\mathbf{G}_{\mathrm{d}}\mathbf{s}_2 + \mathbf{b}_{\mathrm{d}}|^2 \right\|_2. \tag{117}$$

The condition

$$d_{\min}^{(E)} > 0 \tag{118}$$

is equivalent to noiseless symbol identifiability over the finite constellation set. A larger $d_{\min}^{(E)}$ implies stronger robustness against post-detection noise and model mismatch.

The data-phase equivalent measurement matrix can be written in the form

$$\mathbf{G}_{\mathrm{d}} = (\mathbf{I}_{N_t} \otimes \hat{\mathbf{H}})\boldsymbol{\Pi}\mathbf{P}_{\mathrm{tx},d}, \tag{119}$$

where $\boldsymbol{\Pi}$ is a permutation matrix that converts the user-stacked waveform samples into the time-stacked order, and $\mathbf{P}_{\mathrm{tx},d}$ is the data-phase pulse-shaping matrix. Using $\hat{\mathbf{H}} \approx \mathbf{H} = \mathbf{V}\boldsymbol{\Gamma}$, the rank and conditioning of $\mathbf{G}_{\mathrm{d}}$ are governed by the pulse-shaping matrix, the RIS array manifold, and the user channel gains. In particular,

$$\mathrm{rank}(\mathbf{G}_{\mathrm{d}}) \leq \min\left\{N_t\,\mathrm{rank}(\hat{\mathbf{H}}), \mathrm{rank}(\mathbf{P}_{\mathrm{tx},d})\right\}. \tag{120}$$

If $\hat{\mathbf{H}}$ is full column rank and $\mathbf{P}_{\mathrm{tx},d}$ is full column rank, then $\mathbf{G}_{\mathrm{d}}$ is full column rank. Moreover, the smallest singular value satisfies

$$\sigma_{\min}(\mathbf{G}_{\mathrm{d}}) \geq \sigma_{\min}(\hat{\mathbf{H}})\sigma_{\min}(\mathbf{P}_{\mathrm{tx},d}), \tag{121}$$

up to the ordering permutation. Since $\hat{\mathbf{H}} \approx \mathbf{V}\boldsymbol{\Gamma}$, we further have

$$\sigma_{\min}(\hat{\mathbf{H}}) \approx \sigma_{\min}(\mathbf{V}\boldsymbol{\Gamma}) \geq \sigma_{\min}(\mathbf{V}) \min_{l} |\alpha_l|. \tag{122}$$

Therefore, stable symbol detection requires not only reference-wave diversity, but also a well-conditioned pulse-shaping matrix, sufficiently separated RIS steering vectors, and non-negligible user channel gains.

The above analysis indicates that the reference vector $\mathbf{b}_{\mathrm{d}}$ and the effective matrix $\mathbf{G}_{\mathrm{d}}$ play different roles. The matrix $\mathbf{G}_{\mathrm{d}}$ must preserve the differences between constellation points in the complex field domain, while the reference vector $\mathbf{b}_{\mathrm{d}}$ must convert these complex-field differences into distinguishable real-valued intensity patterns. Consequently, reliable reference-assisted holographic detection requires the joint design of the pulse-shaping structure, RIS spatial measurements, pilot/reference phases, and detection algorithms.

D. Energy-Domain Minimum Distance

The identifiability conditions discussed above determine whether two feasible parameters can be distinguished in the absence of noise. However, in practical systems, the intensity measurements are corrupted by post-detection noise and model mismatch. Therefore, beyond noiseless identifiability, it is important to quantify how well different feasible parameters are separated in the real-valued intensity domain. To this end, we introduce the energy-domain minimum distance as a stability metric for reference-assisted phase retrieval.

For two feasible parameters $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X}$, define their energy-domain distance as

$$d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2) = \|\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_1) - \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_2)\|_2, \tag{123}$$

where

$$\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) = |\mathbf{G}\boldsymbol{\theta} + \mathbf{b}|^2 \tag{124}$$

is the noiseless reference-assisted intensity mapping. Equivalently,

$$d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2) = \left\| |\mathbf{G}\boldsymbol{\theta}_1 + \mathbf{b}|^2 - |\mathbf{G}\boldsymbol{\theta}_2 + \mathbf{b}|^2 \right\|_2. \tag{125}$$

The corresponding energy-domain minimum distance over the feasible set $\mathcal{X}$ is defined as

$$d_{\min}^{(E)} = \min_{\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X},\ \boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2} d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2). \tag{126}$$

For finite-alphabet data detection, where $\mathcal{X} = \mathcal{S}^{LK_{\mathrm{d}}}$, the condition

$$d_{\min}^{(E)} > 0 \tag{127}$$

is equivalent to noiseless symbol identifiability over the constellation set. Indeed, if $d_{\min}^{(E)} = 0$, then there exist two distinct symbol vectors that generate the same intensity observation and are therefore indistinguishable. Conversely, if $d_{\min}^{(E)} > 0$, every pair of distinct symbol vectors is separated in the intensity domain, which guarantees unique recovery in the absence of noise.

The energy-domain distance can be further decomposed to reveal the role of the reference wave. Let

$$\Delta\boldsymbol{\theta} = \boldsymbol{\theta}_1 - \boldsymbol{\theta}_2, \quad \bar{\boldsymbol{\theta}} = \boldsymbol{\theta}_1 + \boldsymbol{\theta}_2. \tag{128}$$

Then

$$\begin{aligned} &|\mathbf{G}\boldsymbol{\theta}_1 + \mathbf{b}|^2 - |\mathbf{G}\boldsymbol{\theta}_2 + \mathbf{b}|^2 \\ &= |\mathbf{G}\boldsymbol{\theta}_1|^2 - |\mathbf{G}\boldsymbol{\theta}_2|^2 + 2\,\mathrm{Re}\left\{\mathbf{b}^* \odot \mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)\right\}. \end{aligned} \tag{129}$$

Equivalently, using $\Delta\boldsymbol{\theta}$ and $\bar{\boldsymbol{\theta}}$, we have

$$\begin{aligned} &|\mathbf{G}\boldsymbol{\theta}_1 + \mathbf{b}|^2 - |\mathbf{G}\boldsymbol{\theta}_2 + \mathbf{b}|^2 \\ &= \mathrm{Re}\left\{\left[\mathbf{G}\bar{\boldsymbol{\theta}} + 2\mathbf{b}\right]^* \odot \mathbf{G}\Delta\boldsymbol{\theta}\right\}. \end{aligned} \tag{130}$$

Equation (131) shows that the energy-domain separation is determined by two coupled factors. The term $\mathbf{G}\Delta\boldsymbol{\theta}$ represents the complex-field difference produced by two feasible parameters, while the term $\mathbf{G}\bar{\boldsymbol{\theta}} + 2\mathbf{b}$ determines how this complex-field difference is projected into the real-valued intensity domain. Therefore, the effective matrix $\mathbf{G}$ must preserve parameter differences in the field domain, whereas the reference vector $\mathbf{b}$ must convert these differences into observable energy-domain separations.

For data detection, the energy-domain minimum distance becomes

$$d_{\min,\mathrm{d}}^{(E)} = \min_{\mathbf{s}_1, \mathbf{s}_2 \in \mathcal{S}^{LK_{\mathrm{d}}},\ \mathbf{s}_1 \neq \mathbf{s}_2} \left\| |\mathbf{G}_{\mathrm{d}}\mathbf{s}_1 + \mathbf{b}_{\mathrm{d}}|^2 - |\mathbf{G}_{\mathrm{d}}\mathbf{s}_2 + \mathbf{b}_{\mathrm{d}}|^2 \right\|_2. \tag{131}$$

This metric directly characterizes the robustness of finite-alphabet symbol detection. When the post-detection noise is modeled as an additive Gaussian noise vector with covariance $\sigma_w^2\mathbf{I}$, the pairwise error probability between two candidate symbol vectors can be approximated as

$$P(\mathbf{s}_1 \to \mathbf{s}_2) \approx Q\left(\frac{d_E(\mathbf{s}_1, \mathbf{s}_2)}{2\sigma_w}\right), \tag{132}$$

where $Q(\cdot)$ is the Gaussian tail function. Consequently, a larger $d_{\min,\mathrm{d}}^{(E)}$ generally leads to a lower symbol error probability. This provides a direct link between the reference-assisted identifiability analysis and the detection performance.

The reference vector can therefore be designed to enlarge the energy-domain minimum distance. A natural design criterion is

$$\max_{\mathbf{b}_{\mathrm{d}} \in \mathcal{B}} \quad d_{\min,\mathrm{d}}^{(E)}, \tag{133}$$

where $\mathcal{B}$ denotes the feasible set of reference waves determined by practical amplitude, phase, and hardware constraints. For example, if the reference amplitudes are fixed, i.e., $|b_i| = \rho_i$, the design reduces to the selection of reference phases. Reference phase diversity across RIS elements, time slots, or measurement configurations can increase the probability that every effective symbol difference $\mathbf{G}_{\mathrm{d}}(\mathbf{s}_1 - \mathbf{s}_2)$ is projected onto a nonzero intensity-domain difference.

For continuous channel estimation, the global minimum distance in (127) is generally not suitable because the feasible set is continuous and arbitrarily close parameters can yield arbitrarily small distances. In this case, the stability of the intensity mapping is better characterized locally by the Jacobian introduced in (99). For a small perturbation $\Delta\boldsymbol{\theta}$ around a nominal channel parameter, we have

$$\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta} + \Delta\boldsymbol{\theta}) - \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) = \mathbf{J}(\boldsymbol{\theta})\Delta\bar{\boldsymbol{\theta}} + o(\|\Delta\bar{\boldsymbol{\theta}}\|_2), \tag{134}$$

where

$$\Delta\bar{\boldsymbol{\theta}} = \left[\mathrm{Re}\{\Delta\boldsymbol{\theta}\}^T \quad \mathrm{Im}\{\Delta\boldsymbol{\theta}\}^T\right]^T. \tag{135}$$

Thus, the local energy-domain separation satisfies

$$\|\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta} + \Delta\boldsymbol{\theta}) - \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta})\|_2 \gtrsim \sigma_{\min}(\mathbf{J}(\boldsymbol{\theta}))\|\Delta\bar{\boldsymbol{\theta}}\|_2. \tag{136}$$

Therefore, for channel estimation, increasing $\sigma_{\min}(\mathbf{J}(\boldsymbol{\theta}))$ improves the local stability of the recovery problem.

The above discussion highlights the different roles of the energy-domain distance in the two phases. For finite-alphabet symbol detection, $d_{\min}^{(E)}$ provides a global and computable separability metric over all candidate symbol vectors. For continuous channel estimation, the analogous stability measure is the local singular-value behavior of the Jacobian. In both cases, the reference wave, the pilot or data measurement matrix, and the RIS spatial response jointly determine whether the intensity measurements are not only identifiable but also robust to noise.

IV. Reference Wave Design

The analysis in the previous section shows that the reference vector plays a critical role in converting complex-field differences into real-valued intensity-domain separations. In this section, we discuss how to design the reference wave to improve the identifiability and stability of reference-assisted holographic reception. We focus on the reference phase design, since the reference amplitudes are often constrained by hardware power, dynamic range, and detector linearity.

A. Reference Phase Diversity

Consider the unified reference-assisted intensity model

$$\mathbf{z} = |\mathbf{G}\boldsymbol{\theta} + \mathbf{b}|^2 + \mathbf{w}, \tag{137}$$

where the reference vector is parameterized as

$$\mathbf{b} = \boldsymbol{\rho} \odot e^{j\boldsymbol{\varphi}}, \tag{138}$$

with

$$\boldsymbol{\rho} = [\rho_1, \rho_2, \ldots, \rho_Q]^T, \quad \boldsymbol{\varphi} = [\varphi_1, \varphi_2, \ldots, \varphi_Q]^T. \tag{139}$$

Here, $\rho_q \geq 0$ and $\varphi_q \in [0, 2\pi)$ denote the amplitude and phase of the q-th equivalent reference sample, respectively. The index q may correspond to different RIS elements, time samples, pilot slots, or reference configurations, depending on the considered observation model.

From the energy-domain distance analysis, for two feasible parameters $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$, the difference between their noiseless intensity observations is

$$\begin{aligned} &|\mathbf{G}\boldsymbol{\theta}_1 + \mathbf{b}|^2 - |\mathbf{G}\boldsymbol{\theta}_2 + \mathbf{b}|^2 \\ &= \mathrm{Re}\left\{[\mathbf{G}(\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2) + 2\mathbf{b}]^* \odot \mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)\right\}. \end{aligned} \tag{140}$$

This expression shows that the reference wave affects how a complex-field difference is projected onto the real-valued intensity domain. In particular, the reference-induced part is

$$2\,\mathrm{Re}\left\{\mathbf{b}^* \odot \mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)\right\}. \tag{141}$$

Let

$$[\mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)]_q = \gamma_q e^{j\chi_q}, \quad \gamma_q \geq 0. \tag{142}$$

Then the q-th component of (142) becomes

$$2\rho_q\gamma_q \cos(\chi_q - \varphi_q). \tag{143}$$

Therefore, the reference phase φ_q determines the projection direction of the complex-field difference at the q-th observation. If all reference phases are identical or highly concentrated, certain field differences may be nearly orthogonal to the reference projection direction and thus weakly observable in the intensity domain. Reference phase diversity mitigates this issue by probing different phase directions across observations.

For finite-alphabet data detection, a natural design criterion is to maximize the energy-domain minimum distance:

$$\max_{\boldsymbol{\varphi}} \quad d_{\min}^{(E)}(\boldsymbol{\varphi}) = \min_{\mathbf{s}_1 \neq \mathbf{s}_2} \left\| \left|\mathbf{G}_\mathrm{d}\mathbf{s}_1 + \boldsymbol{\rho} \odot e^{j\boldsymbol{\varphi}}\right|^2 - \left|\mathbf{G}_\mathrm{d}\mathbf{s}_2 + \boldsymbol{\rho} \odot e^{j\boldsymbol{\varphi}}\right|^2 \right\|_2, \tag{144}$$

where $\mathbf{s}_1, \mathbf{s}_2 \in \mathcal{S}^{LK_\mathrm{d}}$. For channel estimation, the feasible set is continuous, and the corresponding local stability criterion is to improve the conditioning of the Jacobian:

$$\max_{\boldsymbol{\varphi}} \quad \sigma_{\min}\left(\mathbf{J}_{\boldsymbol{\varphi}}(\boldsymbol{\theta})\right), \tag{145}$$

or, equivalently, to reduce the condition number of $\mathbf{J}_{\boldsymbol{\varphi}}(\boldsymbol{\theta})$. Here, the subscript $\boldsymbol{\varphi}$ emphasizes that the Jacobian depends on the reference phases through

$$\mathbf{u} = \mathbf{G}\boldsymbol{\theta} + \boldsymbol{\rho} \odot e^{j\boldsymbol{\varphi}}. \tag{146}$$

In practical systems, solving (145) or (146) exactly may be computationally expensive, especially when the constellation dimension or the number of RIS observations is large. Therefore, several structured phase-diversity designs are useful as low-complexity reference patterns.

1) Constant-Phase Reference: The simplest design is to use a common reference phase for all observations:

$$\varphi_q = \varphi_0, \quad q = 1, 2, \ldots, Q. \tag{147}$$

This design is easy to implement but provides limited phase diversity. All complex-field differences are projected along the same reference direction. Consequently, if an effective field difference is nearly quadrature to this direction over many observations, its contribution to the intensity-domain separation becomes weak. Thus, the constant-phase reference serves mainly as a baseline design.

2) Random-Phase Reference: A more robust design is to draw the reference phases independently from a continuous distribution:

$$\varphi_q \sim \mathcal{U}[0, 2\pi), \quad q = 1, 2, \ldots, Q. \tag{148}$$

Random phase diversity reduces the probability that all relevant field differences are simultaneously projected onto weak intensity directions. For a finite set of candidate symbols, if the effective matrix $\mathbf{G}_\mathrm{d}$ preserves all constellation differences, then random continuous phases avoid exact pairwise degeneracies with probability one. This makes random-phase references attractive for validating the identifiability analysis and for providing a simple non-optimized reference design.

3) Uniform Phase-Sweeping Reference: A deterministic alternative is to use uniformly distributed phases:

$$\varphi_q = \varphi_0 + \frac{2\pi(q-1)}{Q}, \quad q = 1, 2, \ldots, Q. \tag{149}$$

or, when multiple reference states are used,

$$\varphi_r = \varphi_0 + \frac{2\pi(r-1)}{R}, \quad r = 1, 2, \ldots, R, \tag{150}$$

where R is the number of reference states. The uniform phase-sweeping design covers the complex plane in a balanced manner and avoids the randomness of (149). It is also hardware-friendly because the phase values can be precomputed and stored as a reference codebook.

To quantify the phase diversity of a given reference pattern, one can use the phase concentration metric

$$\eta_\varphi = \left| \frac{1}{Q_\rho} \sum_{q \in \Omega_\rho} e^{j2\varphi_q} \right|, \tag{151}$$

where

$$\Omega_\rho = \{q : \rho_q > 0\}, \quad Q_\rho = |\Omega_\rho|. \tag{152}$$

The doubled phase $2\varphi_q$ is used because projection directions separated by π are collinear in the real-valued interference term. A value of $\eta_\varphi = 1$ indicates that all reference phases are aligned modulo π, corresponding to poor phase diversity. In contrast, a smaller η_φ indicates a more balanced coverage of projection directions. Random and uniform phase references typically yield a small η_φ when the number of observations is sufficiently large.

The above designs show that reference phase diversity is a practical way to improve the separability of intensity-domain observations without changing the receiver hardware architecture. In the subsequent subsections, this principle will be further used to design reference patterns for pilot-based channel estimation and finite-alphabet symbol detection.

B. Reference Design for Channel Estimation

We next discuss the design of the reference wave for the training phase. Different from finite-alphabet data detection, where the energy-domain minimum distance can be used as a global separability metric, channel estimation involves continuous unknown parameters. Therefore, the reference wave should be designed to improve the local sensitivity and conditioning of the reference-assisted intensity mapping.

Recall the block-level training model

$$\mathbf{z}_{\mathrm{p}} = |\mathbf{G}_{\mathrm{p}}\boldsymbol{\theta}_{\mathrm{c}} + \mathbf{b}_{\mathrm{p}}|^2 + \mathbf{w}_{\mathrm{p}}, \tag{153}$$

where $\boldsymbol{\theta}_{\mathrm{c}} \in \mathbb{C}^{D_{\mathrm{c}}}$ denotes the channel-related unknown parameter, $\mathbf{G}_{\mathrm{p}} \in \mathbb{C}^{Q_{\mathrm{p}}\times D_{\mathrm{c}}}$ is the training measurement matrix, $\mathbf{b}_{\mathrm{p}} \in \mathbb{C}^{Q_{\mathrm{p}}}$ is the training reference vector, and Q_{p} is the total number of training intensity observations. The unknown parameter can be the full vectorized channel $\boldsymbol{\theta}_{\mathrm{c}} = \mathrm{vec}(\mathbf{H})$, or a lower-dimensional structured parameter such as the path-gain vector $\boldsymbol{\alpha}$ when the far-field array manifold is exploited.

Let

$$\mathbf{u}_{\mathrm{p}} = \mathbf{G}_{\mathrm{p}}\boldsymbol{\theta}_{\mathrm{c}} + \mathbf{b}_{\mathrm{p}}. \tag{154}$$

Following (99), the real-valued Jacobian of the training intensity mapping is given by

$$\mathbf{J}_{\mathrm{p}}(\boldsymbol{\theta}_{\mathrm{c}}, \mathbf{b}_{\mathrm{p}}) = 2\left[\mathrm{Re}\left\{\mathrm{diag}(\mathbf{u}_{\mathrm{p}}^*)\mathbf{G}_{\mathrm{p}}\right\}, -\mathrm{Im}\left\{\mathrm{diag}(\mathbf{u}_{\mathrm{p}}^*)\mathbf{G}_{\mathrm{p}}\right\}\right] \in \mathbb{R}^{Q_{\mathrm{p}}\times 2D_{\mathrm{c}}}. \tag{155}$$

A necessary condition for local channel identifiability is $Q_{\mathrm{p}} \geq 2D_{\mathrm{c}}$, and a regular local identifiability condition is

$$\mathrm{rank}\left(\mathbf{J}_{\mathrm{p}}(\boldsymbol{\theta}_{\mathrm{c}}, \mathbf{b}_{\mathrm{p}})\right) = 2D_{\mathrm{c}}. \tag{156}$$

Therefore, the purpose of reference design in the training phase is to make $\mathbf{J}_{\mathrm{p}}$ not only full column rank, but also well conditioned.

Under additive Gaussian post-detection noise with covariance $\sigma_w^2\mathbf{I}$, the Fisher information matrix of the real-valued parameter vector is proportional to

$$\mathbf{F}_{\mathrm{p}} = \frac{1}{\sigma_w^2}\mathbf{J}_{\mathrm{p}}^T\mathbf{J}_{\mathrm{p}}. \tag{157}$$

Thus, a well-designed reference wave should enlarge the smallest singular value of $\mathbf{J}_{\mathrm{p}}$ or reduce the corresponding estimation error bound. Typical design criteria include

$$\max_{\mathbf{b}_{\mathrm{p}}\in\mathcal{B}_{\mathrm{p}}} \quad \sigma_{\min}\left(\mathbf{J}_{\mathrm{p}}(\boldsymbol{\theta}_{\mathrm{c}}, \mathbf{b}_{\mathrm{p}})\right), \tag{158}$$

$$\min_{\mathbf{b}_{\mathrm{p}}\in\mathcal{B}_{\mathrm{p}}} \quad \kappa\left(\mathbf{J}_{\mathrm{p}}(\boldsymbol{\theta}_{\mathrm{c}}, \mathbf{b}_{\mathrm{p}})\right), \tag{159}$$

or

$$\min_{\mathbf{b}_{\mathrm{p}}\in\mathcal{B}_{\mathrm{p}}} \quad \mathrm{tr}\left[\left(\mathbf{J}_{\mathrm{p}}^T\mathbf{J}_{\mathrm{p}} + \epsilon\mathbf{I}\right)^{-1}\right], \tag{160}$$

where $\mathcal{B}_{\mathrm{p}}$ denotes the feasible reference set determined by amplitude, phase, and hardware constraints, and $\epsilon > 0$ is a small regularization parameter.

Since the true channel parameter $\boldsymbol{\theta}_{\mathrm{c}}$ is unknown before training, the above design criteria can be evaluated using a nominal channel parameter, a coarse prior estimate, or a statistical average over a feasible channel set. For example, a robust reference design can be formulated as

$$\max_{\mathbf{b}_{\mathrm{p}}\in\mathcal{B}_{\mathrm{p}}} \quad \min_{\boldsymbol{\theta}_{\mathrm{c}}\in\Theta_{\mathrm{c}}} \sigma_{\min}\left(\mathbf{J}_{\mathrm{p}}(\boldsymbol{\theta}_{\mathrm{c}}, \mathbf{b}_{\mathrm{p}})\right), \tag{161}$$

where Θ_{c} denotes a set of possible channel parameters. Although solving (162) exactly may be difficult, it provides a useful principle: the reference wave should avoid producing poorly conditioned Jacobians over the expected channel region.

The role of the reference wave can be further interpreted from the structure of (156). When the reference is absent, i.e., $\mathbf{b}_{\mathrm{p}} = \mathbf{0}$, the intensity mapping has an inherent global phase ambiguity and the Jacobian loses rank along the infinitesimal phase-rotation direction. With a nonzero reference, the effective field becomes

$$\mathbf{u}_{\mathrm{p}} = \mathbf{G}_{\mathrm{p}}\boldsymbol{\theta}_{\mathrm{c}} + \mathbf{b}_{\mathrm{p}}, \tag{162}$$

and the rows of $\mathbf{J}_{\mathrm{p}}$ are rotated and weighted by the phases and amplitudes of $\mathbf{u}_{\mathrm{p}}$. Therefore, reference phase diversity changes the real-valued projections of the complex training measurements and can remove locally unobservable directions.

Several practical reference designs can be used in the training phase.

1) Random or Uniform Training References: A simple design is to use fixed-amplitude references with random or uniformly swept phases:

$$[\mathbf{b}_{\mathrm{p}}]_q = \rho_q e^{j\varphi_q}, \quad q = 1, 2, \ldots, Q_{\mathrm{p}}. \tag{163}$$

Random phases,

$$\varphi_q \sim \mathcal{U}[0, 2\pi), \tag{164}$$

provide statistically diverse projections and reduce the probability of degenerate phase alignments between the reference wave and the unknown channel-induced field. Alternatively, uniformly swept phases,

$$\varphi_q = \varphi_0 + \frac{2\pi(q-1)}{Q_{\mathrm{p}}}, \tag{165}$$

provide deterministic phase coverage and are easier to implement as a predesigned training reference codebook. These designs are low-complexity and do not require prior channel knowledge.

2) Jacobian-Oriented Reference Selection: If a finite reference codebook is available, the receiver can select the reference pattern that gives the best predicted local conditioning. Let

$$\mathcal{C}_{\mathrm{p}} = \{\mathbf{b}_{\mathrm{p}}^{(1)}, \mathbf{b}_{\mathrm{p}}^{(2)}, \ldots, \mathbf{b}_{\mathrm{p}}^{(C)}\} \tag{166}$$

denote a training reference codebook. Given a nominal channel parameter $\boldsymbol{\theta}_{\mathrm{c}}^{(0)}$, the selected reference can be chosen as

$$\mathbf{b}_{\mathrm{p}}^\star = \arg\max_{\mathbf{b}_{\mathrm{p}}^{(c)}\in\mathcal{C}_{\mathrm{p}}} \sigma_{\min}\left(\mathbf{J}_{\mathrm{p}}(\boldsymbol{\theta}_{\mathrm{c}}^{(0)}, \mathbf{b}_{\mathrm{p}}^{(c)})\right). \tag{167}$$

Alternatively, one may minimize the condition number of the Jacobian:

$$\mathbf{b}_{\mathrm{p}}^\star = \arg\min_{\mathbf{b}_{\mathrm{p}}^{(c)}\in\mathcal{C}_{\mathrm{p}}} \kappa\left(\mathbf{J}_{\mathrm{p}}(\boldsymbol{\theta}_{\mathrm{c}}^{(0)}, \mathbf{b}_{\mathrm{p}}^{(c)})\right). \tag{168}$$

This codebook-based design is suitable for practical systems because the reference candidates can be generated offline and evaluated with low online complexity.

3) Quadrature Reference Probing: When the channel remains approximately constant over multiple training reference states, a stronger design is to use quadrature reference probing. For a given training field $u_q = [\mathbf{G}_{\mathrm{p}}\boldsymbol{\theta}_{\mathrm{c}}]_q$, consider four reference states

$$b_q \in \{\rho,\ -\rho,\ j\rho,\ -j\rho\}. \tag{169}$$

The corresponding intensity differences satisfy

$$\frac{|u_q+\rho|^2-|u_q-\rho|^2}{4\rho} = \mathrm{Re}\{u_q\}, \tag{170}$$

and

$$\frac{|u_q+j\rho|^2-|u_q-j\rho|^2}{4\rho} = \mathrm{Im}\{u_q\}. \tag{171}$$

Thus, by using paired in-phase and quadrature reference states, the receiver can directly extract the real and imaginary components of the effective complex field. This converts the phaseless channel estimation problem into an equivalent linear complex-field estimation problem, at the cost of additional training measurements and stricter coherence requirements.

The quadrature probing design provides a useful performance benchmark for reference-assisted channel estimation. It also shows that the reference wave is not merely an auxiliary power component, but a controllable probing signal that determines how the unknown complex field is encoded into intensity measurements.

In summary, reference design for channel estimation should satisfy two requirements. First, the pilot matrix and the RIS array manifold must make the complex training field sufficiently informative, which is reflected by the rank and conditioning of $\mathbf{G}_{\mathrm{p}}$. Second, the reference phases should provide diverse and nondegenerate real-valued projections of that field, which is reflected by the rank and conditioning of the Jacobian $\mathbf{J}_{\mathrm{p}}$. These two requirements jointly determine the stability and accuracy of the training-stage channel estimation.

C. Reference Design for Symbol Detection

We now consider the reference wave design for the data transmission phase. Different from channel estimation, where the unknown channel parameter is continuous, data detection involves a finite-alphabet symbol vector. Therefore, the reference wave can be designed directly according to the energy-domain separation between different candidate symbol vectors.

Recall the data-phase reference-assisted intensity model

$$\mathbf{z}_{\mathrm{d}} = |\mathbf{G}_{\mathrm{d}}\mathbf{s}_{\mathrm{d}}+\mathbf{b}_{\mathrm{d}}|^2 + \mathbf{w}_{\mathrm{d}}, \tag{172}$$

where $\mathbf{s}_{\mathrm{d}} \in \mathcal{S}^{LK_{\mathrm{d}}}$ is the unknown data-symbol vector, $\mathbf{G}_{\mathrm{d}}$ is the equivalent data-phase measurement matrix, and $\mathbf{b}_{\mathrm{d}}$ denotes the equivalent reference vector in the data phase. For notational simplicity, the dependence of $\mathbf{G}_{\mathrm{d}}$ on the estimated channel is omitted in this subsection.

For two distinct candidate symbol vectors $\mathbf{s}_i, \mathbf{s}_j \in \mathcal{S}^{LK_{\mathrm{d}}}$, define the pairwise energy-domain distance as

$$d_{ij}^{(E)}(\mathbf{b}_{\mathrm{d}}) = \left\| |\mathbf{G}_{\mathrm{d}}\mathbf{s}_i+\mathbf{b}_{\mathrm{d}}|^2 - |\mathbf{G}_{\mathrm{d}}\mathbf{s}_j+\mathbf{b}_{\mathrm{d}}|^2 \right\|_2. \tag{173}$$

The corresponding minimum distance is

$$d_{\min,\mathrm{d}}^{(E)}(\mathbf{b}_{\mathrm{d}}) = \min_{\mathbf{s}_i \neq \mathbf{s}_j} d_{ij}^{(E)}(\mathbf{b}_{\mathrm{d}}). \tag{174}$$

According to the identifiability analysis, $d_{\min,\mathrm{d}}^{(E)}(\mathbf{b}_{\mathrm{d}}) > 0$ guarantees noiseless symbol identifiability, while a larger value improves robustness against post-detection noise. Therefore, a natural reference design criterion for data detection is

$$\max_{\mathbf{b}_{\mathrm{d}} \in \mathcal{B}_{\mathrm{d}}} \quad d_{\min,\mathrm{d}}^{(E)}(\mathbf{b}_{\mathrm{d}}), \tag{175}$$

where $\mathcal{B}_{\mathrm{d}}$ is the feasible set of data-phase reference waves determined by practical amplitude and phase constraints.

To reveal the role of the reference vector, let

$$\Delta\mathbf{s}_{ij} = \mathbf{s}_i - \mathbf{s}_j, \quad \bar{\mathbf{s}}_{ij} = \mathbf{s}_i + \mathbf{s}_j. \tag{176}$$

Then the pairwise intensity difference can be written as

$$\begin{aligned} \boldsymbol{\delta}_{ij}(\mathbf{b}_{\mathrm{d}}) &= |\mathbf{G}_{\mathrm{d}}\mathbf{s}_i+\mathbf{b}_{\mathrm{d}}|^2 - |\mathbf{G}_{\mathrm{d}}\mathbf{s}_j+\mathbf{b}_{\mathrm{d}}|^2 \\ &= \mathrm{Re}\left\{ [\mathbf{G}_{\mathrm{d}}\bar{\mathbf{s}}_{ij}+2\mathbf{b}_{\mathrm{d}}]^* \odot \mathbf{G}_{\mathrm{d}}\Delta\mathbf{s}_{ij} \right\}. \end{aligned} \tag{177}$$

Hence,

$$d_{ij}^{(E)}(\mathbf{b}_{\mathrm{d}}) = \|\boldsymbol{\delta}_{ij}(\mathbf{b}_{\mathrm{d}})\|_2. \tag{178}$$

Equation (178) shows that the reference vector affects the pairwise distance through the interference term

$$2\,\mathrm{Re}\left\{ \mathbf{b}_{\mathrm{d}}^* \odot \mathbf{G}_{\mathrm{d}}\Delta\mathbf{s}_{ij} \right\}. \tag{179}$$

Thus, the reference wave should be selected such that every effective constellation difference $\mathbf{G}_{\mathrm{d}}\Delta\mathbf{s}_{ij}$ produces a sufficiently large real-valued intensity difference.

When the reference amplitudes are fixed, the design variable reduces to the phase vector. Let

$$\mathbf{b}_{\mathrm{d}} = \boldsymbol{\rho}_{\mathrm{d}} \odot e^{j\boldsymbol{\varphi}_{\mathrm{d}}}, \tag{180}$$

where $\boldsymbol{\rho}_{\mathrm{d}}$ is fixed and $\boldsymbol{\varphi}_{\mathrm{d}}$ is to be designed. Then (176) becomes

$$\max_{\boldsymbol{\varphi}_{\mathrm{d}}} \quad \min_{\mathbf{s}_i \neq \mathbf{s}_j} \left\| \left|\mathbf{G}_{\mathrm{d}}\mathbf{s}_i + \boldsymbol{\rho}_{\mathrm{d}} \odot e^{j\boldsymbol{\varphi}_{\mathrm{d}}}\right|^2 - \left|\mathbf{G}_{\mathrm{d}}\mathbf{s}_j + \boldsymbol{\rho}_{\mathrm{d}} \odot e^{j\boldsymbol{\varphi}_{\mathrm{d}}}\right|^2 \right\|_2. \tag{181}$$

This problem is generally nonconvex due to the phase variables and the minimum over all symbol pairs. Moreover, the number of candidate pairs grows rapidly with the constellation dimension. Therefore, low-complexity structured designs are desirable for practical implementation.

1) Constellation-Aware Reference Codebook: A practical approach is to construct a finite reference codebook and select the reference pattern that maximizes the predicted minimum energy-domain distance. Let

$$\mathcal{C}_{\mathrm{d}} = \{\mathbf{b}_{\mathrm{d}}^{(1)}, \mathbf{b}_{\mathrm{d}}^{(2)}, \ldots, \mathbf{b}_{\mathrm{d}}^{(C)}\} \tag{182}$$

be a data-phase reference codebook, where each candidate satisfies the hardware constraints in $\mathcal{B}_{\mathrm{d}}$. The selected reference pattern is

$$\mathbf{b}_{\mathrm{d}}^{\star} = \arg\max_{\mathbf{b}_{\mathrm{d}}^{(c)} \in \mathcal{C}_{\mathrm{d}}} d_{\min,\mathrm{d}}^{(E)}\left(\mathbf{b}_{\mathrm{d}}^{(c)}\right). \tag{183}$$

This design is constellation-aware because the metric is evaluated over the finite symbol set. It is also compatible with practical implementations, since the reference patterns can be generated offline and stored as a reference codebook.

To reduce complexity, the minimum in (175) does not necessarily need to be evaluated over all symbol pairs. In many modulation formats, detection errors are dominated by nearest-neighbor symbol pairs in the effective field domain. Thus, one can define a critical pair set

$$\mathcal{P}_{\mathrm{c}} = \left\{ (i,j) : \left\| \mathbf{G}_{\mathrm{d}}(\mathbf{s}_i - \mathbf{s}_j) \right\|_2 \le \tau \right\}, \tag{184}$$

where τ is a prescribed threshold. Then a reduced-complexity design criterion is

$$\mathbf{b}_{\mathrm{d}}^{\star} = \arg \max_{\mathbf{b}_{\mathrm{d}}^{(c)} \in \mathcal{C}_{\mathrm{d}}} \min_{(i,j)\in\mathcal{P}_{\mathrm{c}}} d_{ij}^{(E)} \left(\mathbf{b}_{\mathrm{d}}^{(c)} \right). \tag{185}$$

This approximation focuses the reference design on the most vulnerable symbol pairs.

2) Random and Uniform Phase References: When no prior optimization is performed, random and uniform phase references provide simple baseline designs. For random phase references, the phases are generated as

$$\varphi_{\mathrm{d},q} \sim \mathcal{U}[0, 2\pi), \quad q = 1, 2, \ldots, Q_{\mathrm{d}}. \tag{186}$$

This design reduces the chance that the effective constellation differences are simultaneously projected onto weak intensity directions. For a finite constellation set, random continuous reference phases avoid exact pairwise degeneracies with probability one, provided that the effective matrix $\mathbf{G}_{\mathrm{d}}$ does not annihilate any valid constellation difference.

A deterministic alternative is the uniform phase reference:

$$\varphi_{\mathrm{d},q} = \varphi_0 + \frac{2\pi(q-1)}{Q_{\mathrm{d}}}, \quad q = 1, 2, \ldots, Q_{\mathrm{d}}. \tag{187}$$

Uniform phase references provide balanced coverage of different projection directions and are easy to implement. They are especially useful when the receiver has many RIS elements or multiple reference states, so that the reference phases can be distributed over the observation dimension.

3) Pairwise-Oriented Phase Refinement: Starting from a random or uniform phase reference, one can further refine the reference phases according to the current bottleneck symbol pairs. Let

$$(i^{\star}, j^{\star}) = \arg \min_{\mathbf{s}_i \neq \mathbf{s}_j} d_{ij}^{(E)}(\mathbf{b}_{\mathrm{d}}) \tag{188}$$

be the current worst-case symbol pair. A simple phase refinement strategy is to adjust $\boldsymbol{\varphi}_{\mathrm{d}}$ to increase

$$\left\| \mathrm{Re}\left\{ \mathbf{b}_{\mathrm{d}}^{*} \odot \mathbf{G}_{\mathrm{d}}(\mathbf{s}_{i^{\star}} - \mathbf{s}_{j^{\star}}) \right\} \right\|_2. \tag{189}$$

For example, if only the reference phases are optimized and the amplitudes are fixed, a favorable phase choice for a single bottleneck pair is to align each reference phase with the corresponding effective difference:

$$\varphi_{\mathrm{d},q} \approx \angle \left[\mathbf{G}_{\mathrm{d}}(\mathbf{s}_{i^{\star}} - \mathbf{s}_{j^{\star}}) \right]_q, \tag{190}$$

or with the opposite direction, since both yield large absolute real projections. In practice, the reference must serve all symbol pairs rather than only one pair, so (191) can be used as a local update or initialization inside a codebook or greedy search procedure.

The reference design for symbol detection can therefore be summarized as a max-min separation problem in the energy domain. The effective matrix $\mathbf{G}_{\mathrm{d}}$ determines the field-domain constellation geometry, while the reference vector $\mathbf{b}_{\mathrm{d}}$ reshapes this geometry after square-law detection. A well-designed reference wave should enlarge the weakest pairwise intensity separation and thereby reduce the probability of symbol confusion under post-detection noise.

D. Joint Reference and Measurement Diversity

The previous subsections focus on the design of the reference vector under a fixed effective measurement matrix. However, the identifiability analysis in Section III shows that the stability of reference-assisted holographic reception is jointly determined by the effective matrix $\mathbf{G}$ and the reference vector $\mathbf{b}$. While the reference wave controls the real-valued projection of the complex field, the measurement matrix determines whether the differences between feasible parameters are preserved before square-law detection. Therefore, additional measurement diversity can further improve channel estimation and data detection.

Suppose that R measurement configurations are used within one training or data block. The r-th configuration is characterized by an effective measurement matrix $\mathbf{G}^{(r)}$ and a reference vector $\mathbf{b}^{(r)}$. The corresponding intensity observation is

$$\mathbf{z}^{(r)} = \left| \mathbf{G}^{(r)} \boldsymbol{\theta} + \mathbf{b}^{(r)} \right|^2 + \mathbf{w}^{(r)}, \quad r = 1, 2, \ldots, R. \tag{191}$$

Stacking all configurations yields

$$\mathbf{z}_{\mathrm{ext}} = \left| \mathbf{G}_{\mathrm{ext}} \boldsymbol{\theta} + \mathbf{b}_{\mathrm{ext}} \right|^2 + \mathbf{w}_{\mathrm{ext}}, \tag{192}$$

where

$$\mathbf{G}_{\mathrm{ext}} = \begin{bmatrix} \mathbf{G}^{(1)} \\ \mathbf{G}^{(2)} \\ \vdots \\ \mathbf{G}^{(R)} \end{bmatrix}, \quad \mathbf{b}_{\mathrm{ext}} = \begin{bmatrix} \mathbf{b}^{(1)} \\ \mathbf{b}^{(2)} \\ \vdots \\ \mathbf{b}^{(R)} \end{bmatrix}. \tag{193}$$

This extended model has the same form as the unified reference-assisted phase retrieval model, but with more intensity observations and richer measurement diversity.

For channel estimation, the extended real-valued Jacobian is obtained by stacking the Jacobians associated with different configurations:

$$\mathbf{J}_{\mathrm{ext}} = \begin{bmatrix} \mathbf{J}^{(1)} \\ \mathbf{J}^{(2)} \\ \vdots \\ \mathbf{J}^{(R)} \end{bmatrix}, \tag{194}$$

where $\mathbf{J}^{(r)}$ is computed from $\mathbf{G}^{(r)}$ and $\mathbf{b}^{(r)}$. The corresponding information matrix is

$$\mathbf{J}_{\mathrm{ext}}^T \mathbf{J}_{\mathrm{ext}} = \sum_{r=1}^{R} \left(\mathbf{J}^{(r)} \right)^T \mathbf{J}^{(r)}. \tag{195}$$

Hence, measurement diversity improves channel estimation when it increases the rank, enlarges the smallest singular value, or reduces the condition number of $\mathbf{J}_{\mathrm{ext}}$. This can be achieved by varying the reference phases, changing the pilot patterns, or switching the RIS measurement configurations.

For finite-alphabet symbol detection, the extended energy-domain minimum distance becomes

$$d_{\min,\mathrm{ext}}^{(E)} = \min_{\mathbf{s}_i \neq \mathbf{s}_j} \left\| \left| \mathbf{G}_{\mathrm{ext}} \mathbf{s}_i + \mathbf{b}_{\mathrm{ext}} \right|^2 - \left| \mathbf{G}_{\mathrm{ext}} \mathbf{s}_j + \mathbf{b}_{\mathrm{ext}} \right|^2 \right\|_2. \tag{196}$$

Compared with a single configuration, multiple configurations can enlarge $d_{\min,\text{ext}}^{(E)}$ by providing additional projections of the same symbol difference. As a result, symbol pairs that are weakly separated under one reference or RIS state may become distinguishable after stacking multiple measurements.

The measurement diversity can be introduced in several ways.

1) Reference-Only Diversity: The simplest approach is to keep the measurement matrix fixed while changing only the reference vector:

$$\mathbf{G}^{(1)} = \mathbf{G}^{(2)} = \cdots = \mathbf{G}^{(R)} = \mathbf{G}, \tag{197}$$

with different reference vectors $\{\mathbf{b}^{(r)}\}_{r=1}^{R}$. This design is attractive when the RIS receiving structure is fixed and only the local reference signal can be controlled. It improves the projection diversity in the intensity domain without changing the physical propagation or spatial sampling structure.

2) RIS Measurement Diversity: If the holographic RIS receiver supports reconfigurable measurement states, the effective matrix can also be varied across configurations. For example, different RIS combining patterns, element weighting states, or spatial sampling modes can lead to different matrices $\{\mathbf{G}^{(r)}\}_{r=1}^{R}$. In this case, measurement diversity improves the field-domain separability before intensity detection. A useful design principle is to choose RIS configurations such that

$$\operatorname{rank}(\mathbf{G}_{\text{ext}}) \tag{198}$$

is increased and

$$\sigma_{\min}(\mathbf{G}_{\text{ext}}) \tag{199}$$

is enlarged. For data detection, the selected configurations should also increase the field-domain separation

$$\|\mathbf{G}_{\text{ext}}(\mathbf{s}_i - \mathbf{s}_j)\|_2 \tag{200}$$

for vulnerable symbol pairs.

3) Joint Reference-RIS Diversity: The most flexible strategy is to jointly vary the measurement matrix and the reference vector:

$$\left\{\mathbf{G}^{(r)}, \mathbf{b}^{(r)}\right\}_{r=1}^{R}. \tag{201}$$

In this case, $\mathbf{G}^{(r)}$ creates diverse complex-field observations, while $\mathbf{b}^{(r)}$ converts these field differences into intensity-domain separations. A general design criterion for data detection is

$$\max_{\{\mathbf{G}^{(r)},\mathbf{b}^{(r)}\}_{r=1}^{R}} \quad d_{\min,\text{ext}}^{(E)}, \tag{202}$$

subject to the hardware constraints of the RIS receiver and the reference-wave generator. For channel estimation, an analogous criterion is

$$\max_{\{\mathbf{G}^{(r)},\mathbf{b}^{(r)}\}_{r=1}^{R}} \quad \sigma_{\min}(\mathbf{J}_{\text{ext}}), \tag{203}$$

or equivalently minimizing the condition number of $\mathbf{J}_{\text{ext}}$.

In practice, the joint design should balance performance improvement and measurement overhead. Increasing the number of configurations provides more intensity observations and usually improves identifiability, but it also increases training duration, reference switching cost, and detection latency. Therefore, a practical design may use a small number of structured configurations, such as uniform reference phase sweeps combined with a limited RIS measurement codebook. The design objective is not necessarily to achieve the globally optimal measurement pattern, but to avoid degenerate projections and to ensure that the weakest channel or symbol differences are sufficiently visible in the intensity domain.

The above discussion provides a unified design guideline for reference-assisted holographic reception. The effective measurement matrix should preserve the relevant differences in the complex field domain, while the reference vector should transform those differences into robust intensity-domain separations. This principle applies to both pilot-based channel estimation and finite-alphabet symbol detection, and it will be validated through the estimation and detection algorithms developed in the following section.

V. Estimation and Detection Algorithms

Based on the reference-assisted observation model and reference-wave design principles developed in the previous sections, this section presents practical estimation and detection algorithms for the proposed holographic RIS receiver. The goal of this section is not to introduce a new family of generic phase retrieval algorithms, but to show how the proposed two-phase communication framework can be implemented using channel estimation and symbol detection procedures that are consistent with the identifiability analysis.

A. Overall Two-Phase Recovery Framework

The proposed receiver operates in two phases: a training phase for channel estimation and a data transmission phase for symbol detection. In the training phase, known pilot symbols and designed reference waves are used to estimate the channel-related parameter. In the data phase, the estimated channel is used to construct the equivalent measurement matrix for detecting the finite-alphabet data symbols from reference-assisted intensity measurements.

The overall recovery procedure is summarized in Algorithm 1. First, the receiver selects or generates a training reference vector $\mathbf{b}_{\text{p}}$ according to the design criteria discussed in Section IV. The users then transmit pilot signals, and the RIS receiver collects the training intensity measurements $\mathbf{z}_{\text{p}}$. Based on the training model

$$\mathbf{z}_{\text{p}} = |\mathbf{G}_{\text{p}}\boldsymbol{\theta}_{\text{c}} + \mathbf{b}_{\text{p}}|^2 + \mathbf{w}_{\text{p}}, \tag{204}$$

the receiver estimates the channel parameter $\boldsymbol{\theta}_{\text{c}}$ and reconstructs the channel estimate $\hat{\mathbf{H}}$.

After channel estimation, the data-phase reference vector $\mathbf{b}_{\text{d}}$ is selected. The receiver then forms the data-phase equivalent measurement matrix $\mathbf{G}_{\text{d}}(\hat{\mathbf{H}})$ according to the estimated channel and the transmit-side pulse-shaping model. Given the data-phase intensity measurements

$$\mathbf{z}_{\text{d}} = \left|\mathbf{G}_{\text{d}}(\hat{\mathbf{H}})\mathbf{s}_{\text{d}} + \mathbf{b}_{\text{d}}\right|^2 + \mathbf{w}_{\text{d}}, \tag{205}$$

the receiver detects the transmitted symbol vector $\mathbf{s}_{\text{d}}$ using either maximum-likelihood detection or a low-complexity phase-retrieval-based detector.

Algorithm 1 Two-Phase Reference-Assisted Holographic Reception

Require: Pilot sequences, constellation set $\mathcal{S}$, RIS geometry, pulse-shaping matrix, reference codebooks or reference design rules.

Ensure: Estimated channel $\hat{\mathbf{H}}$ and detected data symbols $\hat{\mathbf{s}}_{\mathrm{d}}$.

1: **Training phase:**
2: Select the training reference vector $\mathbf{b}_{\mathrm{p}}$ according to the channel-estimation design criterion.
3: Users transmit known pilot signals.
4: RIS receiver collects intensity measurements $\mathbf{z}_{\mathrm{p}}$.
5: Estimate the channel-related parameter $\hat{\boldsymbol{\theta}}_{\mathrm{c}}$ from

$$\mathbf{z}_{\mathrm{p}} = |\mathbf{G}_{\mathrm{p}}\boldsymbol{\theta}_{\mathrm{c}} + \mathbf{b}_{\mathrm{p}}|^2 + \mathbf{w}_{\mathrm{p}}.$$

6: Reconstruct the channel estimate $\hat{\mathbf{H}}$ from $\hat{\boldsymbol{\theta}}_{\mathrm{c}}$.
7: **Data transmission phase:**
8: Select the data-phase reference vector $\mathbf{b}_{\mathrm{d}}$ according to the symbol-detection design criterion.
9: Construct the equivalent data-phase measurement matrix $\mathbf{G}_{\mathrm{d}}(\hat{\mathbf{H}})$.
10: Users transmit data symbols.
11: RIS receiver collects intensity measurements $\mathbf{z}_{\mathrm{d}}$.
12: Detect the data-symbol vector from

$$\mathbf{z}_{\mathrm{d}} = \left|\mathbf{G}_{\mathrm{d}}(\hat{\mathbf{H}})\mathbf{s}_{\mathrm{d}} + \mathbf{b}_{\mathrm{d}}\right|^2 + \mathbf{w}_{\mathrm{d}}.$$

13: Output $\hat{\mathbf{H}}$ and $\hat{\mathbf{s}}_{\mathrm{d}}$.

The two phases share the same reference-assisted phase retrieval structure, but they differ in the feasible set of the unknown variable. In the training phase, the unknown channel parameter is continuous, and the recovery performance is closely related to the rank and conditioning of the Jacobian of the intensity mapping. In the data phase, the unknown symbol vector belongs to a finite constellation set, and the detection performance is governed by the energy-domain minimum distance. Therefore, the subsequent subsections present tailored algorithms for channel estimation and symbol detection, respectively.

B. Reference-Assisted Channel Estimation

We first consider the channel estimation problem in the training phase. Given the designed training reference vector $\mathbf{b}_{\mathrm{p}}$ and the known pilot matrix, the receiver estimates the channel-related parameter from reference-assisted intensity measurements. According to the training model, the observation can be written as

$$\mathbf{z}_{\mathrm{p}} = |\mathbf{G}_{\mathrm{p}}\boldsymbol{\theta}_{\mathrm{c}} + \mathbf{b}_{\mathrm{p}}|^2 + \mathbf{w}_{\mathrm{p}}, \tag{206}$$

where $\boldsymbol{\theta}_{\mathrm{c}} \in \mathbb{C}^{D_{\mathrm{c}}}$ denotes the channel-related unknown parameter. For full channel estimation, $\boldsymbol{\theta}_{\mathrm{c}} = \mathrm{vec}(\mathbf{H})$. When the far-field structure $\mathbf{H} = \mathbf{V}\boldsymbol{\Gamma}$ is exploited and the angles of arrival are known or pre-estimated, $\boldsymbol{\theta}_{\mathrm{c}}$ can be reduced to the complex path-gain vector $\boldsymbol{\alpha}$.

The channel estimation problem is formulated as a nonlinear least-squares problem:

$$\hat{\boldsymbol{\theta}}_{\mathrm{c}} = \arg\min_{\boldsymbol{\theta}_{\mathrm{c}}\in\mathbb{C}^{D_{\mathrm{c}}}} \left\|\mathbf{z}_{\mathrm{p}} - |\mathbf{G}_{\mathrm{p}}\boldsymbol{\theta}_{\mathrm{c}} + \mathbf{b}_{\mathrm{p}}|^2\right\|_2^2. \tag{207}$$

Since the objective function is nonconvex due to the square-law operation, we solve (208) using a damped Gauss–Newton method. This choice is consistent with the local identifiability analysis in Section III, where the Jacobian of the intensity mapping plays a central role.

Let

$$\mathbf{u}_{\mathrm{p}}^{(t)} = \mathbf{G}_{\mathrm{p}}\boldsymbol{\theta}_{\mathrm{c}}^{(t)} + \mathbf{b}_{\mathrm{p}} \tag{208}$$

be the predicted complex field at the t-th iteration, and define the predicted intensity as

$$\hat{\mathbf{z}}_{\mathrm{p}}^{(t)} = \left|\mathbf{u}_{\mathrm{p}}^{(t)}\right|^2. \tag{209}$$

The residual vector is

$$\mathbf{e}^{(t)} = \mathbf{z}_{\mathrm{p}} - \hat{\mathbf{z}}_{\mathrm{p}}^{(t)}. \tag{210}$$

To apply the Gauss–Newton update, we represent the complex channel parameter by its real-valued form:

$$\bar{\boldsymbol{\theta}}_{\mathrm{c}} = \left[\mathrm{Re}\{\boldsymbol{\theta}_{\mathrm{c}}\}^T \quad \mathrm{Im}\{\boldsymbol{\theta}_{\mathrm{c}}\}^T\right]^T \in \mathbb{R}^{2D_{\mathrm{c}}}. \tag{211}$$

The real-valued Jacobian at the t-th iteration is

$$\mathbf{J}_{\mathrm{p}}^{(t)} = 2\left[\mathrm{Re}\left\{\mathrm{diag}\left((\mathbf{u}_{\mathrm{p}}^{(t)})^*\right)\mathbf{G}_{\mathrm{p}}\right\}, -\mathrm{Im}\left\{\mathrm{diag}\left((\mathbf{u}_{\mathrm{p}}^{(t)})^*\right)\mathbf{G}_{\mathrm{p}}\right\}\right]. \tag{212}$$

Then the damped Gauss–Newton increment is obtained by solving

$$\Delta\bar{\boldsymbol{\theta}}_{\mathrm{c}}^{(t)} = \left[(\mathbf{J}_{\mathrm{p}}^{(t)})^T\mathbf{J}_{\mathrm{p}}^{(t)} + \lambda^{(t)}\mathbf{I}\right]^{-1}(\mathbf{J}_{\mathrm{p}}^{(t)})^T\mathbf{e}^{(t)}, \tag{213}$$

where $\lambda^{(t)} \geq 0$ is a damping parameter used to improve numerical stability. The real-valued channel parameter is updated as

$$\bar{\boldsymbol{\theta}}_{\mathrm{c}}^{(t+1)} = \bar{\boldsymbol{\theta}}_{\mathrm{c}}^{(t)} + \Delta\bar{\boldsymbol{\theta}}_{\mathrm{c}}^{(t)}. \tag{214}$$

After convergence, the complex channel parameter $\hat{\boldsymbol{\theta}}_{\mathrm{c}}$ is reconstructed from $\bar{\boldsymbol{\theta}}_{\mathrm{c}}$.

A practical initialization can be obtained from a reference-dominated linear approximation. When the reference wave is sufficiently strong compared with the unknown field, the quadratic term $|\mathbf{G}_{\mathrm{p}}\boldsymbol{\theta}_{\mathrm{c}}|^2$ can be ignored in the early initialization stage, leading to

$$\mathbf{z}_{\mathrm{p}} - |\mathbf{b}_{\mathrm{p}}|^2 \approx 2\,\mathrm{Re}\left\{\mathbf{b}_{\mathrm{p}}^* \odot \mathbf{G}_{\mathrm{p}}\boldsymbol{\theta}_{\mathrm{c}}\right\}. \tag{215}$$

This real-valued linear model can be solved by least squares to provide an initial estimate for the iterative refinement. Alternatively, random initialization or a coarse channel prior can be used when such information is available.

The complete reference-assisted channel estimation procedure is summarized in Algorithm 2.

The complexity of each Gauss–Newton iteration is mainly determined by forming the Jacobian and solving the linear system in (214). For a channel parameter dimension D_{c} and Q_{p} training intensity observations, the per-iteration complexity is approximately

$$\mathcal{O}\left(Q_{\mathrm{p}}D_{\mathrm{c}}^2 + D_{\mathrm{c}}^3\right). \tag{216}$$

Algorithm 2 Reference-Assisted Channel Estimation

Require: Training intensity measurements $\mathbf{z}_{\mathrm{p}}$, training matrix $\mathbf{G}_{\mathrm{p}}$, reference vector $\mathbf{b}_{\mathrm{p}}$, maximum iteration number I_{c}, and stopping threshold ϵ_{c}.

Ensure: Estimated channel parameter $\hat{\boldsymbol{\theta}}_{\mathrm{c}}$ and reconstructed channel $\hat{\mathbf{H}}$.

1: Initialize $\boldsymbol{\theta}_{\mathrm{c}}^{(0)}$ using (216) or a coarse prior.
2: **for** $t = 0, 1, \ldots, I_{\mathrm{c}} - 1$ **do**
3: Compute
$$\mathbf{u}_{\mathrm{p}}^{(t)} = \mathbf{G}_{\mathrm{p}}\boldsymbol{\theta}_{\mathrm{c}}^{(t)} + \mathbf{b}_{\mathrm{p}}.$$
4: Compute the residual
$$\mathbf{e}^{(t)} = \mathbf{z}_{\mathrm{p}} - |\mathbf{u}_{\mathrm{p}}^{(t)}|^2.$$
5: Construct the Jacobian $\mathbf{J}_{\mathrm{p}}^{(t)}$ according to (213).
6: Compute the damped Gauss–Newton increment from (214).
7: Update $\bar{\boldsymbol{\theta}}_{\mathrm{c}}^{(t+1)}$ according to (215).
8: **if** $\|\Delta\bar{\boldsymbol{\theta}}_{\mathrm{c}}^{(t)}\|_2 / \|\bar{\boldsymbol{\theta}}_{\mathrm{c}}^{(t)}\|_2 < \epsilon_{\mathrm{c}}$ **then**
9: **break**
10: **end if**
11: **end for**
12: Reconstruct $\hat{\boldsymbol{\theta}}_{\mathrm{c}}$ from its real-valued representation.
13: Reconstruct $\hat{\mathbf{H}}$ from $\hat{\boldsymbol{\theta}}_{\mathrm{c}}$.

Thus, the total complexity is

$$\mathcal{O}\left(I_{\mathrm{c}}\left(Q_{\mathrm{p}}D_{\mathrm{c}}^2 + D_{\mathrm{c}}^3\right)\right), \tag{217}$$

where I_{c} is the number of iterations. This complexity highlights the benefit of exploiting the structured far-field channel model. If the full channel vector $\mathrm{vec}(\mathbf{H})$ is estimated, then $D_{\mathrm{c}} = N_{\mathrm{r}}L$. In contrast, if the angles of arrival are known or pre-estimated and only the complex path gains are estimated, then $D_{\mathrm{c}} = L$, which significantly reduces the computational cost.

The estimated channel $\hat{\mathbf{H}}$ is then used to construct the data-phase measurement matrix $\mathbf{G}_{\mathrm{d}}(\hat{\mathbf{H}})$ for symbol detection. The impact of the channel estimation error on the subsequent data detection will be analyzed later in this section.

C. Reference-Assisted Symbol Detection

We next consider the data-symbol detection problem in the data transmission phase. After the training phase, the receiver obtains the channel estimate $\hat{\mathbf{H}}$ and constructs the data-phase equivalent measurement matrix $\mathbf{G}_{\mathrm{d}}(\hat{\mathbf{H}})$. For notational simplicity, we denote it by $\hat{\mathbf{G}}_{\mathrm{d}}$ in this subsection. The data-phase intensity observation is given by

$$\mathbf{z}_{\mathrm{d}} = \left|\hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s}_{\mathrm{d}} + \mathbf{b}_{\mathrm{d}}\right|^2 + \mathbf{w}_{\mathrm{d}}, \tag{218}$$

where $\mathbf{s}_{\mathrm{d}} \in \mathcal{S}^{D_{\mathrm{s}}}$ is the unknown data-symbol vector, $D_{\mathrm{s}} = LK_{\mathrm{d}}$ is the total number of data symbols in the considered block, $\mathbf{b}_{\mathrm{d}}$ is the designed data-phase reference vector, and $\mathbf{w}_{\mathrm{d}}$ is the effective post-detection noise.

1) Maximum-Likelihood Detection: When the post-detection noise is modeled as additive white Gaussian noise in the intensity domain, the maximum-likelihood detector is equivalent to the following least-squares detector:

$$\hat{\mathbf{s}}_{\mathrm{ML}} = \arg\min_{\mathbf{s}\in\mathcal{S}^{D_{\mathrm{s}}}} \left\|\mathbf{z}_{\mathrm{d}} - \left|\hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s} + \mathbf{b}_{\mathrm{d}}\right|^2\right\|_2^2. \tag{219}$$

The ML detector directly searches over the finite constellation set and therefore provides a natural performance benchmark for reference-assisted symbol detection. Its reliability is closely related to the energy-domain minimum distance defined in Section III. Specifically, if the designed reference vector enlarges

$$d_{\min,\mathrm{d}}^{(E)} = \min_{\mathbf{s}_i\neq\mathbf{s}_j} \left\|\left|\hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s}_i + \mathbf{b}_{\mathrm{d}}\right|^2 - \left|\hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s}_j + \mathbf{b}_{\mathrm{d}}\right|^2\right\|_2, \tag{220}$$

then the pairwise separation between candidate symbols is improved, leading to a lower probability of symbol confusion.

However, the complexity of exhaustive ML detection grows exponentially with the number of detected symbols. For a constellation size $|\mathcal{S}|$, the search space has cardinality $|\mathcal{S}|^{D_{\mathrm{s}}}$. The corresponding complexity is approximately

$$\mathcal{O}\left(|\mathcal{S}|^{D_{\mathrm{s}}}Q_{\mathrm{d}}\right), \tag{221}$$

where Q_{d} is the number of data-phase intensity observations. Therefore, ML detection is mainly suitable for short symbol blocks or for benchmarking lower-complexity detectors.

2) Projected Gauss–Newton Detection: To reduce the detection complexity, we also consider a projected Gauss–Newton detector. The main idea is to first relax the finite-alphabet constraint and solve a continuous nonlinear least-squares problem:

$$\hat{\mathbf{s}}_{\mathrm{cont}} = \arg\min_{\mathbf{s}\in\mathbb{C}^{D_{\mathrm{s}}}} \left\|\mathbf{z}_{\mathrm{d}} - \left|\hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s} + \mathbf{b}_{\mathrm{d}}\right|^2\right\|_2^2, \tag{222}$$

and then project the obtained continuous estimate onto the constellation set.

Let

$$\mathbf{u}_{\mathrm{d}}^{(t)} = \hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s}^{(t)} + \mathbf{b}_{\mathrm{d}} \tag{223}$$

be the predicted complex field at the t-th iteration. The predicted intensity and residual are

$$\hat{\mathbf{z}}_{\mathrm{d}}^{(t)} = |\mathbf{u}_{\mathrm{d}}^{(t)}|^2, \tag{224}$$

and

$$\mathbf{e}_{\mathrm{d}}^{(t)} = \mathbf{z}_{\mathrm{d}} - \hat{\mathbf{z}}_{\mathrm{d}}^{(t)}. \tag{225}$$

Similar to the channel estimation stage, define the real-valued symbol vector as

$$\bar{\mathbf{s}} = \left[\mathrm{Re}\{\mathbf{s}\}^T \quad \mathrm{Im}\{\mathbf{s}\}^T\right]^T \in \mathbb{R}^{2D_{\mathrm{s}}}. \tag{226}$$

The corresponding real-valued Jacobian is

$$\mathbf{J}_{\mathrm{d}}^{(t)} = 2\Big[\mathrm{Re}\Big\{\mathrm{diag}\big((\mathbf{u}_{\mathrm{d}}^{(t)})^*\big)\hat{\mathbf{G}}_{\mathrm{d}}\Big\},\ -\mathrm{Im}\Big\{\mathrm{diag}\big((\mathbf{u}_{\mathrm{d}}^{(t)})^*\big)\hat{\mathbf{G}}_{\mathrm{d}}\Big\}\Big]. \tag{227}$$

The damped Gauss–Newton increment is computed as

$$\Delta\bar{\mathbf{s}}^{(t)} = \left[(\mathbf{J}_{\mathrm{d}}^{(t)})^T\mathbf{J}_{\mathrm{d}}^{(t)} + \mu^{(t)}\mathbf{I}\right]^{-1}(\mathbf{J}_{\mathrm{d}}^{(t)})^T\mathbf{e}_{\mathrm{d}}^{(t)}, \tag{228}$$

Algorithm 3 Projected Gauss–Newton Symbol Detection

Require: Data intensity measurements $\mathbf{z}_\mathrm{d}$, equivalent matrix $\hat{\mathbf{G}}_\mathrm{d}$, reference vector $\mathbf{b}_\mathrm{d}$, constellation set $\mathcal{S}$, maximum iteration number I_d, and stopping threshold ϵ_d.

Ensure: Detected symbol vector $\hat{\mathbf{s}}_\mathrm{d}$.

Initialize $\mathbf{s}^{(0)}$ randomly, from a linearized estimate, or from the previous detected block.
for $t = 0, 1, \ldots, I_\mathrm{d} - 1$ **do**
    Compute
$$\mathbf{u}_\mathrm{d}^{(t)} = \hat{\mathbf{G}}_\mathrm{d}\mathbf{s}^{(t)} + \mathbf{b}_\mathrm{d}.$$
    Compute the residual
$$\mathbf{e}_\mathrm{d}^{(t)} = \mathbf{z}_\mathrm{d} - |\mathbf{u}_\mathrm{d}^{(t)}|^2.$$
    Construct the Jacobian $\mathbf{J}_\mathrm{d}^{(t)}$ according to (228).
    Compute the damped Gauss–Newton increment from (229).
    Update the continuous estimate according to (230).
    Optionally project $\mathbf{s}^{(t+1)}$ onto $\mathcal{S}^{D_\mathrm{s}}$ using (231).
    if $\|\Delta\bar{\mathbf{s}}^{(t)}\|_2/\|\bar{\mathbf{s}}^{(t)}\|_2 < \epsilon_\mathrm{d}$ **then**
        break
    end if
end for
Output
$$\hat{\mathbf{s}}_\mathrm{d} = \mathcal{Q}_\mathcal{S}\left(\mathbf{s}^{(t+1)}\right).$$

where $\mu^{(t)} \geq 0$ is a damping factor. The continuous estimate is updated by

$$\bar{\mathbf{s}}^{(t+1)} = \bar{\mathbf{s}}^{(t)} + \Delta\bar{\mathbf{s}}^{(t)}. \tag{229}$$

After each iteration, or after convergence, the continuous estimate can be projected onto the constellation set:

$$\mathbf{s}^{(t+1)} \leftarrow \mathcal{Q}_\mathcal{S}\left(\mathbf{s}^{(t+1)}\right), \tag{230}$$

where $\mathcal{Q}_\mathcal{S}(\cdot)$ denotes element-wise nearest-neighbor quantization onto $\mathcal{S}$. If the projection is applied after each iteration, the algorithm becomes a projected Gauss–Newton detector. If the projection is applied only after convergence, the algorithm first performs continuous phase retrieval and then makes a final constellation decision.

The projected Gauss–Newton detector is summarized in Algorithm 3.

3) Biased Gerchberg–Saxton Detection: As another low-complexity alternative, a biased Gerchberg–Saxton detector can be used by exploiting the known reference vector. At the t-th iteration, the current field estimate is

$$\mathbf{u}_\mathrm{d}^{(t)} = \hat{\mathbf{G}}_\mathrm{d}\mathbf{s}^{(t)} + \mathbf{b}_\mathrm{d}. \tag{231}$$

The magnitude constraint imposed by the intensity measurements is enforced as

$$\tilde{\mathbf{u}}_\mathrm{d}^{(t)} = \sqrt{\mathbf{z}_\mathrm{d}} \odot e^{j\angle \mathbf{u}_\mathrm{d}^{(t)}}. \tag{232}$$

Then the unknown symbol vector is updated by removing the reference vector and projecting back through the measurement matrix:

$$\mathbf{s}^{(t+1)} = \hat{\mathbf{G}}_\mathrm{d}^{\dagger}\left(\tilde{\mathbf{u}}_\mathrm{d}^{(t)} - \mathbf{b}_\mathrm{d}\right), \tag{233}$$

where $\hat{\mathbf{G}}_\mathrm{d}^{\dagger}$ denotes the Moore–Penrose pseudo-inverse of $\hat{\mathbf{G}}_\mathrm{d}$. Finally, the updated estimate is projected onto the constellation set:

$$\mathbf{s}^{(t+1)} \leftarrow \mathcal{Q}_\mathcal{S}\left(\mathbf{s}^{(t+1)}\right). \tag{234}$$

This method is simple and computationally efficient when the pseudo-inverse can be precomputed. However, its performance depends on the conditioning of $\hat{\mathbf{G}}_\mathrm{d}$ and on the quality of the reference-assisted measurements.

4) Complexity Discussion: The complexity of ML detection is exponential in the number of detected symbols, as shown in (222). In contrast, the projected Gauss–Newton detector has per-iteration complexity

$$\mathcal{O}\left(Q_\mathrm{d}D_\mathrm{s}^2 + D_\mathrm{s}^3\right), \tag{235}$$

which comes from forming the Jacobian and solving the damped normal equation. The biased Gerchberg–Saxton detector has lower per-iteration complexity. If $\hat{\mathbf{G}}_\mathrm{d}^{\dagger}$ is precomputed, each iteration mainly requires matrix-vector multiplications, with complexity approximately

$$\mathcal{O}\left(Q_\mathrm{d}D_\mathrm{s}\right). \tag{236}$$

Therefore, ML detection is suitable as a benchmark for small-scale systems, while projected Gauss–Newton and biased Gerchberg–Saxton detectors provide scalable alternatives for larger data blocks.

The choice of detector also depends on the reference-wave design. A larger energy-domain minimum distance improves the reliability of ML detection and provides a more favorable objective landscape for iterative detectors. Thus, the reference design in Section IV and the detection algorithms in this section are complementary: the former improves the separability of the intensity-domain observations, while the latter recovers the transmitted symbols from these observations.

D. Impact of Channel Estimation Error and Complexity Analysis

The preceding symbol detection algorithms assume that the data-phase equivalent measurement matrix is constructed from the estimated channel. Therefore, the channel estimation error inevitably introduces model mismatch in the data detection stage. In this subsection, we analyze the impact of this mismatch and summarize the computational complexity of the proposed two-phase recovery procedure.

Let the estimated channel be expressed as

$$\hat{\mathbf{H}} = \mathbf{H} + \Delta\mathbf{H}, \tag{237}$$

where $\Delta\mathbf{H}$ denotes the channel estimation error. Accordingly, the data-phase equivalent measurement matrix can be written as

$$\hat{\mathbf{G}}_\mathrm{d} = \mathbf{G}_\mathrm{d} + \Delta\mathbf{G}_\mathrm{d}, \tag{238}$$

where $\mathbf{G}_{\mathrm{d}}$ is the ideal measurement matrix constructed from the true channel, and $\Delta\mathbf{G}_{\mathrm{d}}$ is the corresponding perturbation induced by $\Delta\mathbf{H}$.

The true data-phase intensity observation is

$$\mathbf{z}_{\mathrm{d}} = |\mathbf{G}_{\mathrm{d}}\mathbf{s}_{\mathrm{d}} + \mathbf{b}_{\mathrm{d}}|^2 + \mathbf{w}_{\mathrm{d}}. \tag{239}$$

However, the detector evaluates candidate symbols using the mismatched model

$$\hat{\mathcal{T}}_{\mathrm{d}}(\mathbf{s}) = \left|\hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s} + \mathbf{b}_{\mathrm{d}}\right|^2. \tag{240}$$

For a given candidate symbol vector $\mathbf{s}$, the intensity-domain mismatch caused by channel estimation error is

$$\Delta\mathbf{z}_{\mathrm{CSI}}(\mathbf{s}) = |(\mathbf{G}_{\mathrm{d}} + \Delta\mathbf{G}_{\mathrm{d}})\mathbf{s} + \mathbf{b}_{\mathrm{d}}|^2 - |\mathbf{G}_{\mathrm{d}}\mathbf{s} + \mathbf{b}_{\mathrm{d}}|^2. \tag{241}$$

When $\Delta\mathbf{G}_{\mathrm{d}}$ is small, the first-order approximation gives

$$\Delta\mathbf{z}_{\mathrm{CSI}}(\mathbf{s}) \approx 2\,\mathrm{Re}\left\{(\mathbf{G}_{\mathrm{d}}\mathbf{s} + \mathbf{b}_{\mathrm{d}})^* \odot \Delta\mathbf{G}_{\mathrm{d}}\mathbf{s}\right\}. \tag{242}$$

This expression shows that channel estimation error acts as an additional intensity-domain perturbation. Its impact depends not only on the channel error magnitude, but also on the reference-assisted field $\mathbf{G}_{\mathrm{d}}\mathbf{s} + \mathbf{b}_{\mathrm{d}}$ and the symbol-dependent perturbation $\Delta\mathbf{G}_{\mathrm{d}}\mathbf{s}$.

The robustness of data detection under channel estimation error can be interpreted through the energy-domain distance. For two candidate symbol vectors $\mathbf{s}_i$ and $\mathbf{s}_j$, define the ideal pairwise intensity separation as

$$d_{ij}^{(E)} = \left\| |\mathbf{G}_{\mathrm{d}}\mathbf{s}_i + \mathbf{b}_{\mathrm{d}}|^2 - |\mathbf{G}_{\mathrm{d}}\mathbf{s}_j + \mathbf{b}_{\mathrm{d}}|^2 \right\|_2. \tag{243}$$

Due to post-detection noise and channel estimation error, the effective perturbation in the data detection stage can be approximated as

$$\mathbf{e}_{\mathrm{eff}} = \mathbf{w}_{\mathrm{d}} + \Delta\mathbf{z}_{\mathrm{CSI}}(\mathbf{s}_{\mathrm{d}}). \tag{244}$$

A sufficient condition for correctly distinguishing $\mathbf{s}_{\mathrm{d}}$ from another candidate $\mathbf{s}'$ is

$$d_E(\mathbf{s}_{\mathrm{d}}, \mathbf{s}') > 2\|\mathbf{e}_{\mathrm{eff}}\|_2. \tag{245}$$

Therefore, a larger energy-domain minimum distance provides higher tolerance to both post-detection noise and channel estimation error. This observation further justifies the reference design criterion in Section IV, which aims to enlarge $d_{\min,\mathrm{d}}^{(E)}$ for finite-alphabet symbol detection.

The channel estimation quality can be evaluated by the normalized mean-square error (NMSE)

$$\mathrm{NMSE}_{\mathrm{H}} = \frac{\mathbb{E}\left\{\|\hat{\mathbf{H}} - \mathbf{H}\|_F^2\right\}}{\mathbb{E}\left\{\|\mathbf{H}\|_F^2\right\}}. \tag{246}$$

In the simulations, we will compare the data detection performance under perfect CSI and estimated CSI to quantify the effect of channel estimation error on the end-to-end system performance.

Finally, we summarize the computational complexity of the main estimation and detection procedures. For reference-assisted channel estimation using damped Gauss–Newton iterations, the per-iteration complexity is dominated by forming the Jacobian and solving the damped normal equation. For a channel parameter dimension D_{c} and Q_{p} training observations, the total complexity is approximately

$$\mathcal{O}\left(I_{\mathrm{c}}\left(Q_{\mathrm{p}}D_{\mathrm{c}}^2 + D_{\mathrm{c}}^3\right)\right), \tag{247}$$

where I_{c} is the number of channel-estimation iterations. If the full channel is estimated, then $D_{\mathrm{c}} = N_{\mathrm{r}}L$. If the far-field structure is exploited and only the complex path gains are estimated, then $D_{\mathrm{c}} = L$, resulting in a much lower computational cost.

For data detection, exhaustive ML detection has complexity

$$\mathcal{O}\left(|\mathcal{S}|^{D_{\mathrm{s}}}Q_{\mathrm{d}}\right), \tag{248}$$

where $D_{\mathrm{s}} = LK_{\mathrm{d}}$ is the number of detected data symbols and Q_{d} is the number of data-phase intensity observations. The projected Gauss–Newton detector has total complexity

$$\mathcal{O}\left(I_{\mathrm{d}}\left(Q_{\mathrm{d}}D_{\mathrm{s}}^2 + D_{\mathrm{s}}^3\right)\right), \tag{249}$$

where I_{d} is the number of detection iterations. The biased Gerchberg–Saxton detector has lower complexity. When the pseudo-inverse of $\hat{\mathbf{G}}_{\mathrm{d}}$ is precomputed, each iteration mainly involves matrix-vector multiplications, leading to

$$\mathcal{O}\left(I_{\mathrm{d}}Q_{\mathrm{d}}D_{\mathrm{s}}\right). \tag{250}$$

The above analysis shows that the proposed two-phase receiver can trade off detection performance and computational complexity. ML detection provides a benchmark for small-scale symbol blocks, while projected Gauss–Newton and biased Gerchberg–Saxton detectors provide scalable alternatives. Moreover, exploiting the structured far-field channel model significantly reduces the channel-estimation dimension. Together with the reference-wave design in Section IV, these algorithms provide a practical implementation of the proposed reference-assisted holographic RIS receiver.

VI. Simulation Results

In this section, numerical simulations are conducted to evaluate the proposed reference-assisted holographic RIS receiver. We first examine the channel estimation performance in the training phase, then investigate the symbol detection performance under both perfect and estimated channel state information (CSI). Finally, we study the impact of reference-wave design and RIS measurement diversity on the energy-domain separability and detection reliability.

A. Simulation Setup

We consider an uplink multi-user communication system assisted by a holographic RIS receiver. Unless otherwise specified, the RIS receiver is modeled as a uniform planar array placed on the yz-plane with $N_y \times N_z$ receiving elements. The total number of RIS elements is

$$N_{\mathrm{r}} = N_y N_z. \tag{251}$$

The inter-element spacings are set as

$$d_y = d_z = \frac{\lambda}{2}, \tag{252}$$

where λ is the carrier wavelength. The carrier frequency is denoted by f_c, and the corresponding wavelength is $\lambda = c/f_c$, where c is the speed of light.

The far-field channel model in Section II is adopted. The channel matrix is generated as

$$\mathbf{H} = \mathbf{V}\mathbf{\Gamma}, \tag{253}$$

where $\mathbf{V}$ is the RIS array manifold matrix and $\mathbf{\Gamma} = \mathrm{diag}(\alpha_1, \ldots, \alpha_L)$ contains the complex channel gains of the L users. The channel gains are modeled as

$$\alpha_l = \sqrt{\beta_l} e^{j\omega_l}, \tag{254}$$

where β_l denotes the large-scale channel power gain and ω_l is uniformly distributed over $[0, 2\pi)$. The users' angles of arrival are selected according to the considered simulation scenario. In particular, to study the effect of spatial correlation, the angular separation between users will be varied in some simulations.

For the transmitter, each user transmits symbols drawn from a normalized constellation $\mathcal{S}$, such as QPSK or 16-QAM. The pulse-shaping matrix is constructed according to the pulse-shaping filter introduced in Section II. Unless otherwise specified, the data block contains K_d symbols per user, and the total data-symbol dimension is

$$D_\mathrm{s} = LK_\mathrm{d}. \tag{255}$$

During the training phase, each user transmits known pilot symbols over T_p pilot observations.

The received intensity measurements are generated according to the reference-assisted square-law model

$$\mathbf{z} = |\mathbf{G}\boldsymbol{\theta} + \mathbf{b}|^2 + \mathbf{w}, \tag{256}$$

where $\mathbf{w}$ is modeled as additive real-valued Gaussian noise in the intensity domain. The signal-to-noise ratio (SNR) is defined as

$$\mathrm{SNR} = 10\log_{10}\left(\frac{\mathbb{E}\left\{\left\||\mathbf{G}\boldsymbol{\theta} + \mathbf{b}|^2\right\|_2^2\right\}}{\mathbb{E}\{\|\mathbf{w}\|_2^2\}}\right). \tag{257}$$

We compare the following reference-wave designs:

- **No reference**: $\mathbf{b} = \mathbf{0}$.
- **Constant reference**: all reference entries have the same phase.
- **Random reference**: the reference phases are independently drawn from $\mathcal{U}[0, 2\pi)$.
- **Uniform reference**: the reference phases are uniformly distributed over $[0, 2\pi)$.
- **Codebook-optimized reference**: the reference pattern is selected from a finite codebook according to the design criteria in Section IV.

For fair comparison, the total reference power is kept identical among different reference designs unless otherwise stated.

For channel estimation, the damped Gauss–Newton algorithm in Section V is used. The estimation performance is measured by the normalized mean-square error (NMSE)

$$\mathrm{NMSE}_\mathrm{H} = \frac{\mathbb{E}\left\{\|\hat{\mathbf{H}} - \mathbf{H}\|_F^2\right\}}{\mathbb{E}\{\|\mathbf{H}\|_F^2\}}. \tag{258}$$

TABLE I: Default Simulation Parameters

Parameter	Default value
Carrier frequency f_c	3.5 GHz
RIS array size	$N_y \times N_z = 8 \times 8$
Number of RIS elements N_r	64
Element spacing	$d_y = d_z = \lambda/2$
Number of users L	2 or 4
Modulation constellation $\mathcal{S}$	QPSK / 16-QAM
Pilot length T_p	scenario-dependent
Data block length K_d	scenario-dependent
Reference amplitude	fixed unless otherwise stated
Reference phase designs	none / constant / random / uniform / codebook
Channel estimator	damped Gauss–Newton
Symbol detectors	ML / projected GN / biased GS
Performance metrics	NMSE / SER / $d_\mathrm{min}^{(E)}$

For symbol detection, we compare maximum-likelihood (ML) detection, projected Gauss–Newton detection, and biased Gerchberg–Saxton (GS) detection. The detection performance is evaluated using the symbol error rate (SER), defined as

$$\mathrm{SER} = \frac{\text{number of incorrectly detected symbols}}{\text{total number of transmitted symbols}}. \tag{259}$$

To connect the simulations with the theoretical analysis, we also evaluate the energy-domain minimum distance

$$d_\mathrm{min}^{(E)} = \min_{\mathbf{s}_i \neq \mathbf{s}_j} \left\| |\mathbf{G}_\mathrm{d}\mathbf{s}_i + \mathbf{b}_\mathrm{d}|^2 - |\mathbf{G}_\mathrm{d}\mathbf{s}_j + \mathbf{b}_\mathrm{d}|^2 \right\|_2, \tag{260}$$

as well as the singular-value and condition-number metrics associated with the effective measurement matrix and the Jacobian. All results are averaged over independent channel realizations, noise realizations, and transmitted data blocks.

The default simulation parameters are summarized in Table I. Unless otherwise stated, these default values are used in the following simulations.

B. Channel Estimation Performance

In this subsection, we evaluate the performance of the proposed reference-assisted channel estimation scheme in the training phase. The main objective is to verify the role of the reference wave in improving the local identifiability and stability of channel estimation. Unless otherwise specified, the default simulation parameters in Table I are adopted.

We first compare the channel estimation accuracy under different reference-wave designs. The NMSE performance versus SNR is shown in Fig. **??**. The compared schemes include no reference, constant-phase reference, random-phase reference, uniform-phase reference, and codebook-optimized reference. This comparison is used to verify whether reference phase diversity can improve the conditioning of the intensity-based channel estimation problem.

Next, we study the effect of the pilot length on channel estimation. According to the local identifiability analysis, the number of intensity observations should be sufficiently large to make the Jacobian full column rank and well conditioned. Fig. **??** shows the NMSE as a function of the pilot length T_p. This result is intended to demonstrate how additional pilot observations improve the stability of reference-assisted channel estimation.

We then evaluate the influence of reference power. The reference wave provides phase anchoring through the interference term, but its amplitude also affects the dynamic range and the sensitivity of the square-law measurements. Fig. **??** presents the NMSE performance under different reference amplitudes. This figure is used to determine a reasonable reference-power region for the proposed receiver.

To connect the simulation results with the Jacobian-based analysis, we also evaluate the singular-value behavior of the real-valued Jacobian. Fig. **??** shows the minimum singular value and/or condition number of the Jacobian under different reference-wave designs. This result is used to verify that reference patterns leading to larger $\sigma_{\min}(\mathbf{J}_{\mathrm{p}})$ or smaller $\kappa(\mathbf{J}_{\mathrm{p}})$ also lead to improved NMSE performance.

Finally, we examine the effect of user angular separation. For the far-field channel model, the RIS array manifold matrix $\mathbf{V}$ becomes ill-conditioned when the users have close angles of arrival. This reduces the separability of the channel parameters and degrades the channel estimation performance. Fig. **??** shows the NMSE versus the angular separation between users. The corresponding condition number of the array manifold matrix can also be plotted to illustrate the relationship between spatial correlation and estimation accuracy.

Overall, these results are expected to validate the theoretical insights in Sections III and IV: the training reference wave improves channel estimation by enhancing the local sensitivity of the intensity mapping, while the RIS array manifold and pilot structure determine whether the channel parameters are sufficiently distinguishable in the complex field domain.

C. Symbol Detection Performance with Perfect CSI

In this subsection, we evaluate the data-symbol detection performance under perfect CSI. The purpose is to isolate the impact of reference-assisted intensity measurements and detection algorithms from channel estimation errors. Therefore, the data-phase equivalent measurement matrix $\mathbf{G}_{\mathrm{d}}$ is constructed using the true channel matrix $\mathbf{H}$. Unless otherwise specified, the default simulation parameters in Table I are adopted.

We first compare different symbol detection algorithms under the same reference-wave design. Fig. **??** shows the SER versus SNR for maximum-likelihood (ML) detection, projected Gauss–Newton (GN) detection, and biased Gerchberg–Saxton (GS) detection. The ML detector serves as a benchmark, while projected GN and biased GS provide lower-complexity alternatives. This comparison evaluates the performance-complexity tradeoff of the proposed reference-assisted detection framework.

Next, we investigate the impact of reference-wave design on symbol detection. Fig. **??** compares the SER performance of different reference structures, including no reference, constant-phase reference, random-phase reference, uniform-phase reference, and codebook-optimized reference. For a fair comparison, the total reference power is kept the same for all reference-assisted schemes. This figure is intended to verify that reference phase diversity improves the separability of finite-alphabet symbols in the intensity domain.

To directly validate the energy-domain minimum distance analysis, Fig. **??** presents $d_{\min}^{(E)}$ under different reference-wave designs. According to the theoretical analysis, a larger $d_{\min}^{(E)}$ implies stronger pairwise separability after square-law detection. This result is used to explain the SER trends observed in Fig. **??**.

We further examine the relationship between the energy-domain minimum distance and the detection performance. Fig. **??** plots the SER as a function of $d_{\min}^{(E)}$ by varying the reference pattern or reference phase codebook. This figure is used to verify that improving the energy-domain minimum distance leads to a lower symbol error rate, which supports the max-min reference design criterion introduced in Section IV.

Finally, we study the effect of modulation order on the proposed detection framework. Fig. **??** compares the SER performance for different constellations, such as QPSK and 16-QAM. Higher-order modulation leads to denser constellation points and smaller effective pairwise separations, making symbol detection more sensitive to intensity-domain ambiguity and post-detection noise.

Overall, the perfect-CSI results are expected to demonstrate that the proposed reference-assisted receiver can reliably recover finite-alphabet symbols from intensity-only measurements when the reference wave is properly designed. They also show that the energy-domain minimum distance provides a meaningful indicator for predicting and improving symbol detection performance.

D. End-to-End Symbol Detection with Estimated CSI

In this subsection, we evaluate the end-to-end data detection performance when the channel is estimated from the training phase. Different from the perfect-CSI results in the previous subsection, the data-phase equivalent measurement matrix is now constructed from the estimated channel $\hat{\mathbf{H}}$. Therefore, the detection performance is affected by both post-detection noise and channel estimation error. Unless otherwise specified, the same reference design is used consistently in the training and data phases, and the default simulation parameters in Table I are adopted.

We first compare the SER performance under perfect CSI and estimated CSI. Fig. **??** shows the SER versus SNR for the proposed receiver using different symbol detectors. The perfect-CSI curves provide performance benchmarks, while the estimated-CSI curves quantify the degradation caused by channel estimation error. This comparison validates the necessity of accurate reference-assisted channel estimation for reliable data detection.

Next, we investigate the impact of pilot length on the end-to-end detection performance. Fig. **??** shows the SER as a function of the pilot length T_{p}. A larger pilot length generally improves the channel estimate, which in turn reduces the mismatch in the data-phase measurement matrix. This result links the channel estimation accuracy in the training phase to the final symbol detection reliability.

To further quantify the influence of channel estimation error, Fig. **??** plots the SER as a function of the channel estimation NMSE. This can be obtained either by using different pilot lengths and training SNRs or by injecting controlled channel perturbations into the true channel. The purpose of this figure is to verify that a larger channel estimation error leads to a stronger intensity-domain model mismatch and thus degrades symbol detection performance.

We also compare the end-to-end performance under different training reference designs. Fig. **??** shows the SER obtained with no reference, constant-phase reference, random-phase reference, uniform-phase reference, and codebook-optimized reference in the training phase. This comparison demonstrates how the training reference design affects the final data detection performance through the quality of the estimated channel.

Finally, we evaluate the impact of using different reference designs in the data phase under estimated CSI. Fig. **??** compares the SER performance of different data-phase references while using the same channel estimation procedure. This result separates the effect of data-phase symbol separability from the effect of training-phase channel estimation accuracy.

Overall, these results are expected to show that the proposed reference-assisted receiver remains effective under estimated CSI, but its performance depends on both the training-stage channel estimation quality and the data-stage energy-domain separability. They also verify the analysis in Section V: channel estimation error introduces additional intensity-domain mismatch, while a larger energy-domain minimum distance provides better robustness to this mismatch.

E. Effect of Reference and RIS Measurement Diversity

In this subsection, we further investigate how reference diversity and RIS measurement diversity affect the performance of the proposed receiver. The purpose is to verify the design principle developed in Section IV: the effective measurement matrix should preserve the relevant complex-field differences, while the reference wave should project these differences into distinguishable intensity-domain observations. Unless otherwise specified, the default simulation parameters in Table I are adopted.

We first study the impact of the number of RIS receiving elements. Fig. **??** shows the energy-domain minimum distance $d_{\min}^{(E)}$ as a function of the number of RIS elements N_{r}. As the number of RIS elements increases, the receiver obtains more spatial intensity observations, which generally improves the separability of different symbol vectors in the energy domain.

The corresponding SER performance is shown in Fig. **??**. This figure is used to verify that the improvement in $d_{\min}^{(E)}$ obtained from additional RIS observations can be translated into lower symbol detection error rates. Together with Fig. **??**, this result highlights the role of the holographic RIS as a spatial measurement device rather than merely a passive aperture.

Next, we evaluate the effect of multiple reference states. Fig. **??** plots $d_{\min}^{(E)}$ versus the number of reference states R. Each reference state provides an additional intensity projection of the same underlying complex field. Therefore, increasing R improves measurement diversity and can reduce the probability that a vulnerable symbol pair remains weakly separated under all reference states.

Fig. **??** further shows the SER performance under different numbers of reference states. This result is intended to demonstrate the tradeoff between detection reliability and measurement overhead. Although more reference states provide stronger intensity-domain diversity, they also require additional measurement time or reference switching operations.

We then examine the effect of user angular separation. For the far-field RIS channel model, smaller angular separation leads to more correlated steering vectors and a more ill-conditioned array manifold matrix $\mathbf{V}$. Fig. **??** shows the condition number or minimum singular value of $\mathbf{V}$ as a function of the user angular separation. This figure directly illustrates how the spatial geometry affects the conditioning of the RIS measurement matrix.

The corresponding end-to-end detection performance is shown in Fig. **??**. When the users have close angles of arrival, the steering vectors become highly correlated, which reduces the effective field-domain separation and degrades both channel estimation and symbol detection. As the angular separation increases, the RIS array manifold becomes better conditioned, leading to improved detection reliability.

Finally, we compare reference-only diversity and joint reference-RIS measurement diversity. In the reference-only case, the effective measurement matrix is fixed while the reference phases are varied. In the joint diversity case, both the reference vector and the RIS measurement configuration are varied across measurement states. Fig. **??** compares the resulting $d_{\min}^{(E)}$ and/or SER performance. This comparison is used to verify that varying both $\mathbf{G}$ and $\mathbf{b}$ provides richer observations than changing the reference vector alone.

Overall, these results are expected to confirm that the proposed receiver benefits from both reference diversity and RIS spatial measurement diversity. The reference wave improves the intensity-domain projection of complex-field differences, while the RIS array and its measurement configurations determine whether these differences are sufficiently preserved before square-law detection. This validates the central design principle of the proposed reference-assisted holographic RIS receiver.

VII. CONCLUSION

This paper presented a comprehensive framework for signal recovery in holographic RIS receivers using only amplitude measurements. By employing a known reference wave and formulating the recovery as a nonlinear least-squares problem, we enable the estimation of both amplitude and phase of the transmitted signal. The proposed Gauss-Newton algorithm efficiently solves this optimization, with simulations demonstrating accurate recovery under various SNR conditions.

The mathematical derivation provides complete details from signal transmission through the channel, interference with the reference wave at the RIS, intensity measurement, to the final recovery algorithm. This framework serves as a foundation for more advanced RIS receiver designs, including those with adaptive reflection coefficients, multiple users, and frequency-selective channels.

Future work will extend this approach to practical implementations with hardware impairments, investigate deep learning-based recovery for reduced complexity, and explore applications in joint communication and sensing with RIS.

Appendix A

Appendix B

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